CardamomOT — analysis report

/Users/eventre/Documents/GitHub/CardamomOT/experimental_datasets/Schiebinger

Generated on 2026-10-19 16:47

Run

Split	train
Stimulus penalisation	1.0
Prior penalisation	1.0
Genes	100
Stimuli	2
Observed cells	5413
Timepoints	0, 12, 24, 36, 48, 60, 72, 84, 96, 108, 120, 132, 144, 156, 168, 180, 192, 204, 216, 228, 240, 252, 264, 276, 288, 300, 312, 324, 336, 348, 360, 372, 384, 396, 408, 420, 432
Cell types	Neural, Stromal, Trophoblast, iPSC, Epithelial, MEF
Network shown	inter_simul.npy
Embeddings	UMAP; mRNA: log1p=True, normalise=False, cell_depth_for_representation=True (counts / depth factor, model draws taken at the depth of the cells they mimic)

Perturbations (perturbation_simulation)

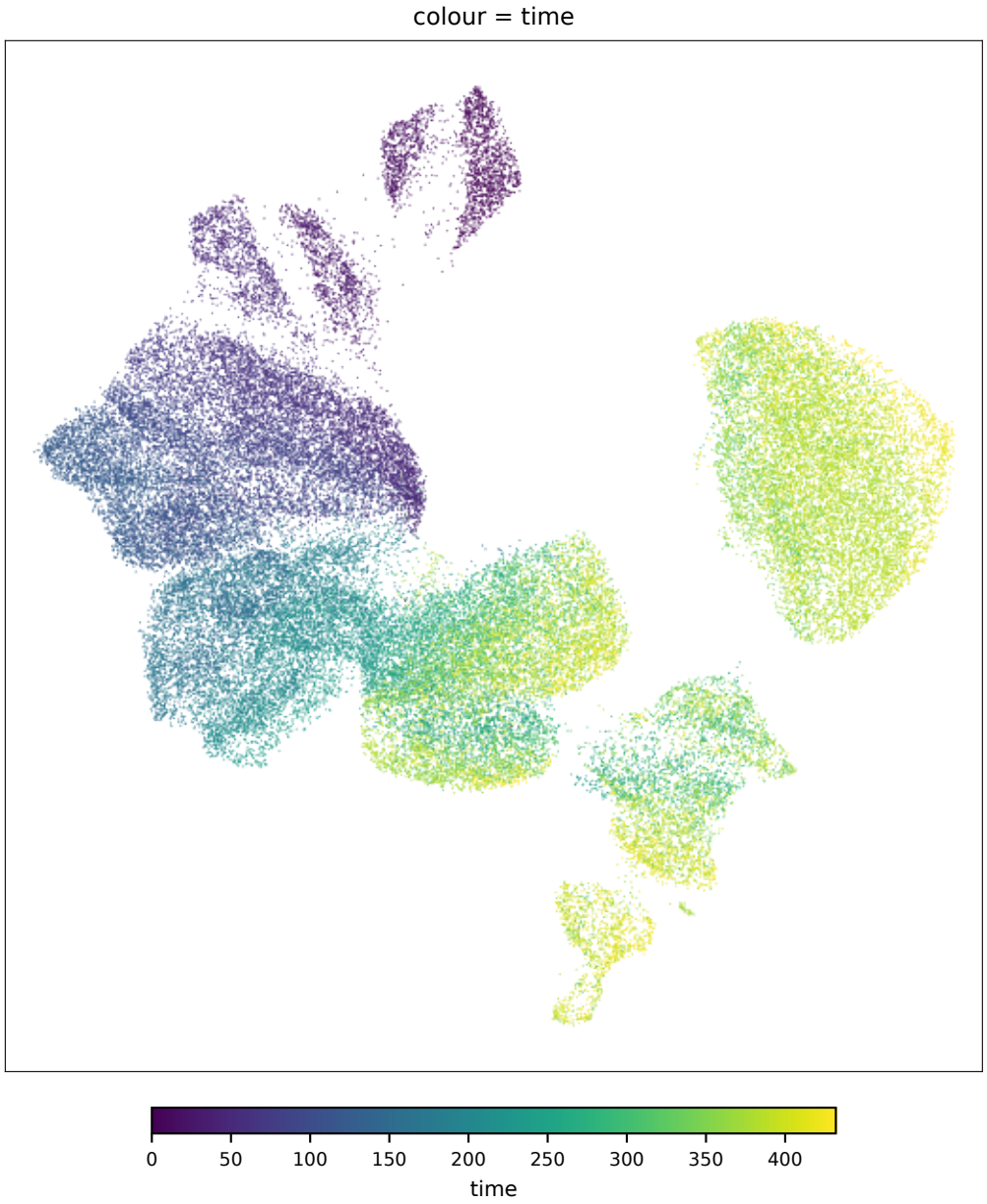
- ✓ P1 OV Obox6
- ✓ P2 OV Zfp42

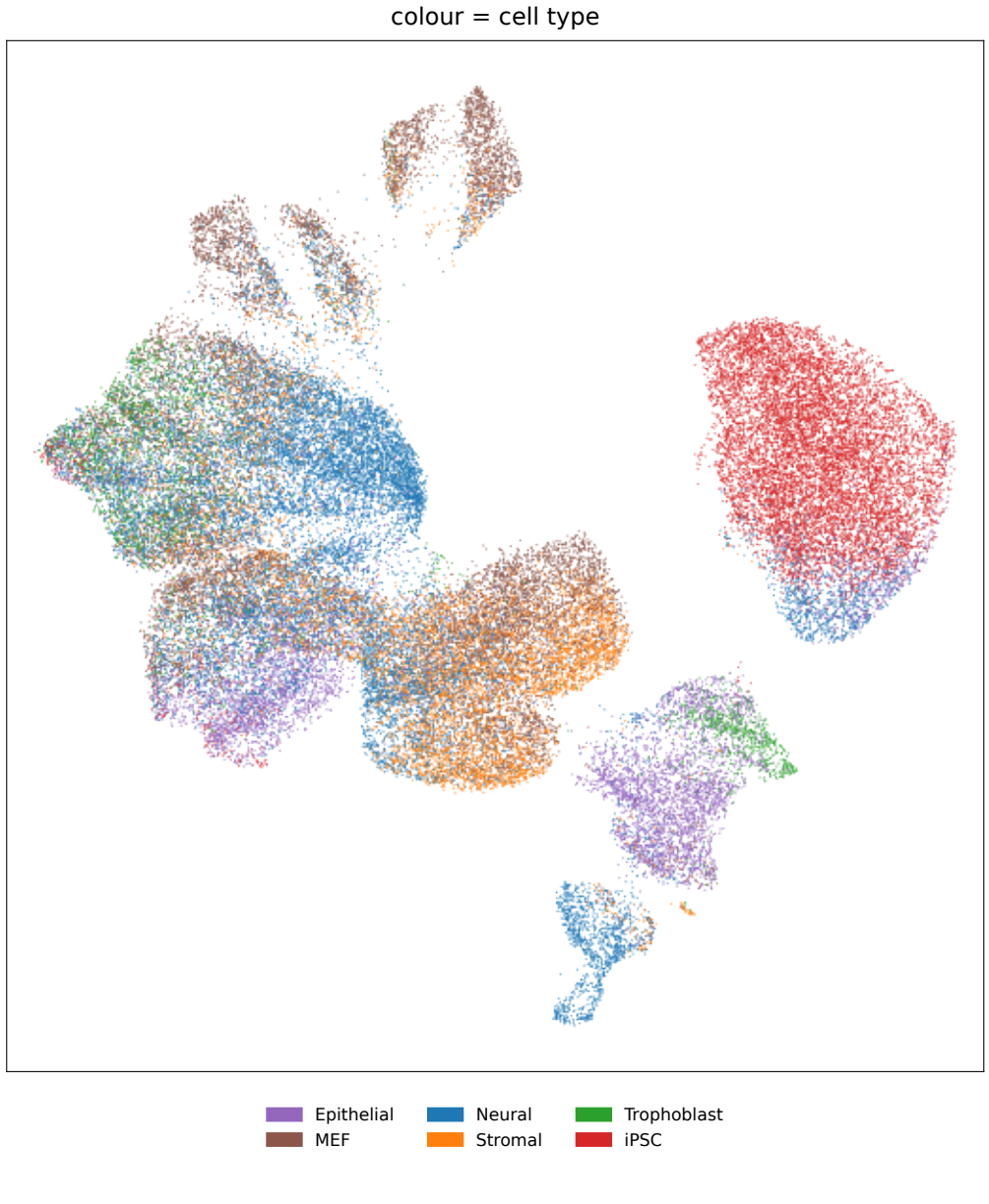
Contents

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2. Gene regulatory network — regulatory power (violin plots), top-20 regulators + stimulus and top-20 regulated genes (mutual top-10 edges)
3. In-silico perturbations — overview across KO/OV, then one page per perturbation
4. Proliferation — prior vs learned net rates, population growth, proteins driving growth; protein dilution at the birth rate
5. Learned dynamics — mRNA and protein fields (mechanistic velocity, displacement along trajectories), summary on mRNA
6. Held-out test cells — predictions with the network fixed vs the test data

Data — UMAP of all the cells

Data/data.h5ad, 50897 cells: log1p(counts), 2000 highly variable genes, PCA (50 components), kNN graph, UMAP. Stored in cardamomOT/embedding_data.npz.

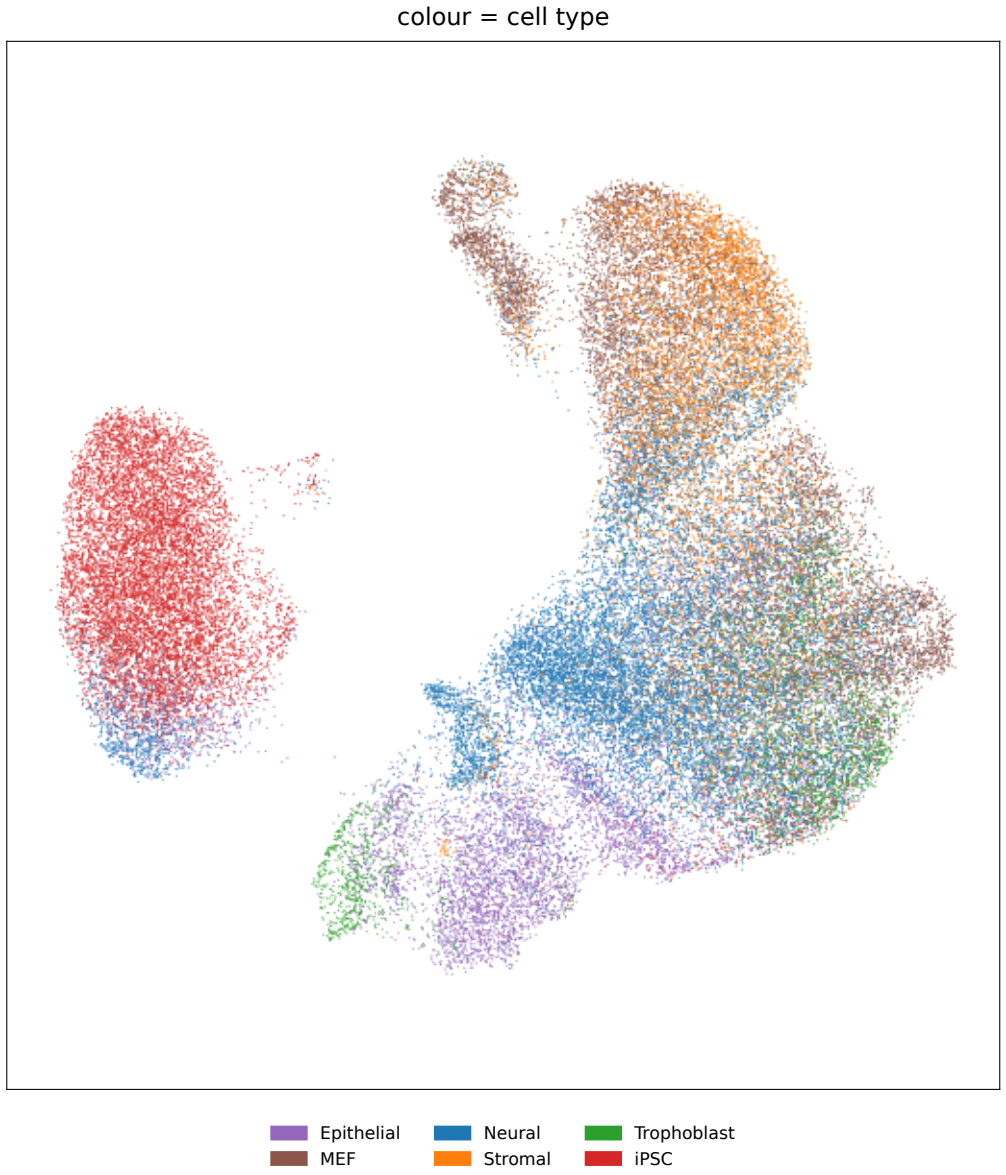




Data — UMAP of all the cells, genes of the gene selection

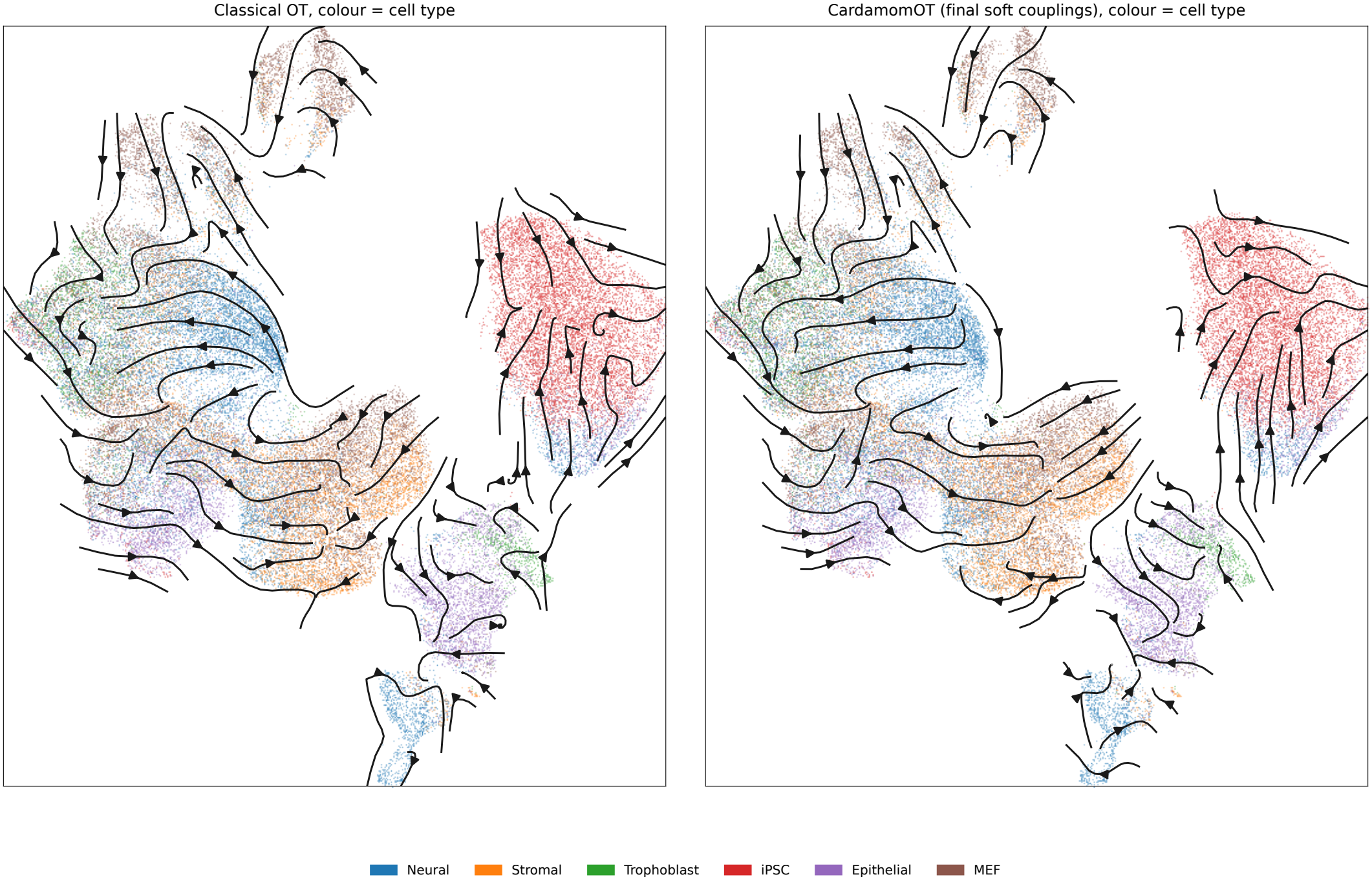
Data/data.h5ad, 50897 cells: log1p(counts), the 100 genes of the model (select_genes), PCA (50 components), kNN graph, UMAP. Stored in cardamomOT/embedding_data_genes.npz.





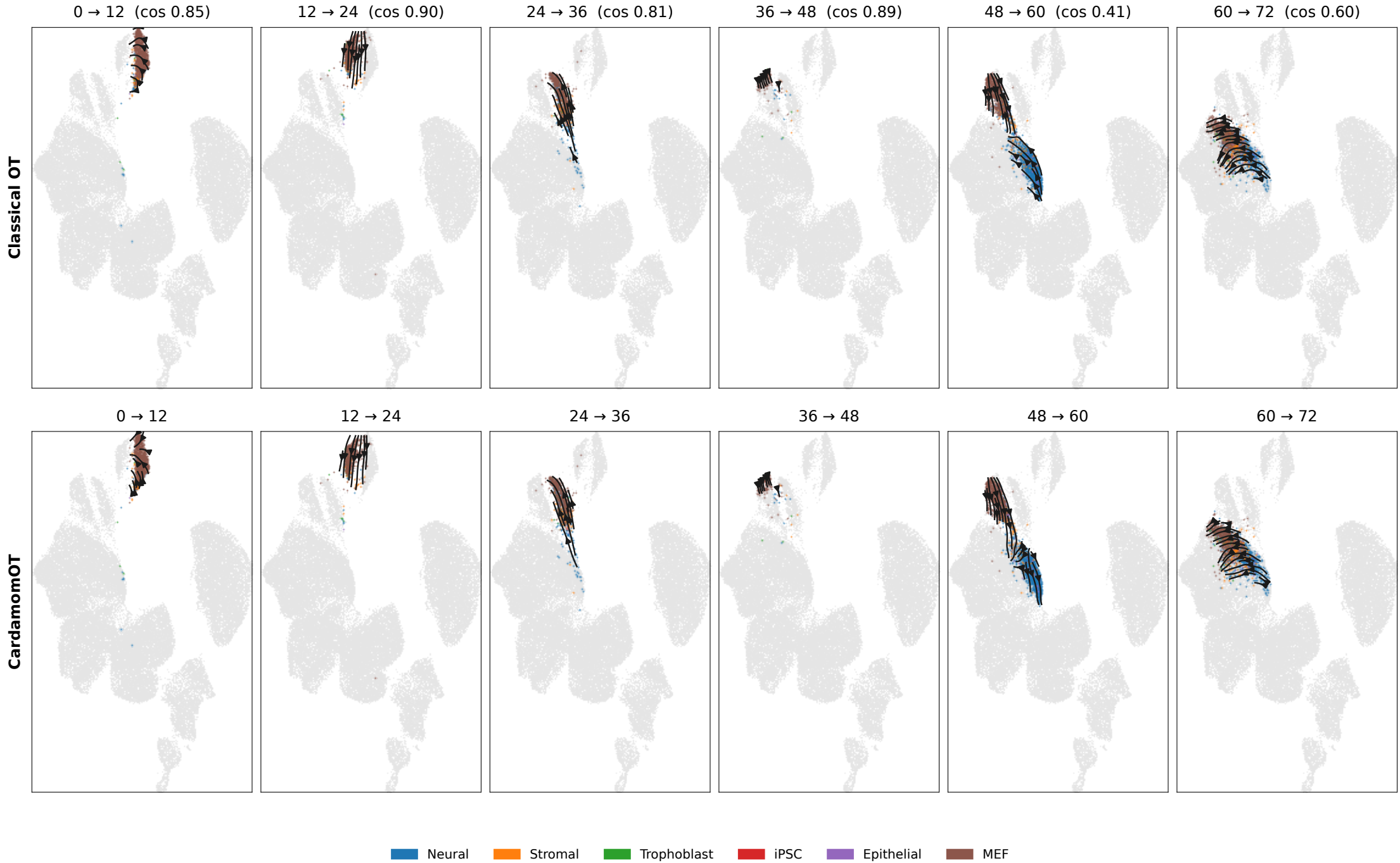
Teaser — displacement fields of classical OT and CardamomOT

Displacement to the mean descendant (x(t+1) − x(t), not divided by Δt), all times. Agreement on the cells with both direct displacements: weighted cosine 0.63 (PCA space). CardamomOT: final soft couplings of the trajectories (5413 cells); cells never reached or outside the training set: Gaussian kernel on the model mRNAs.



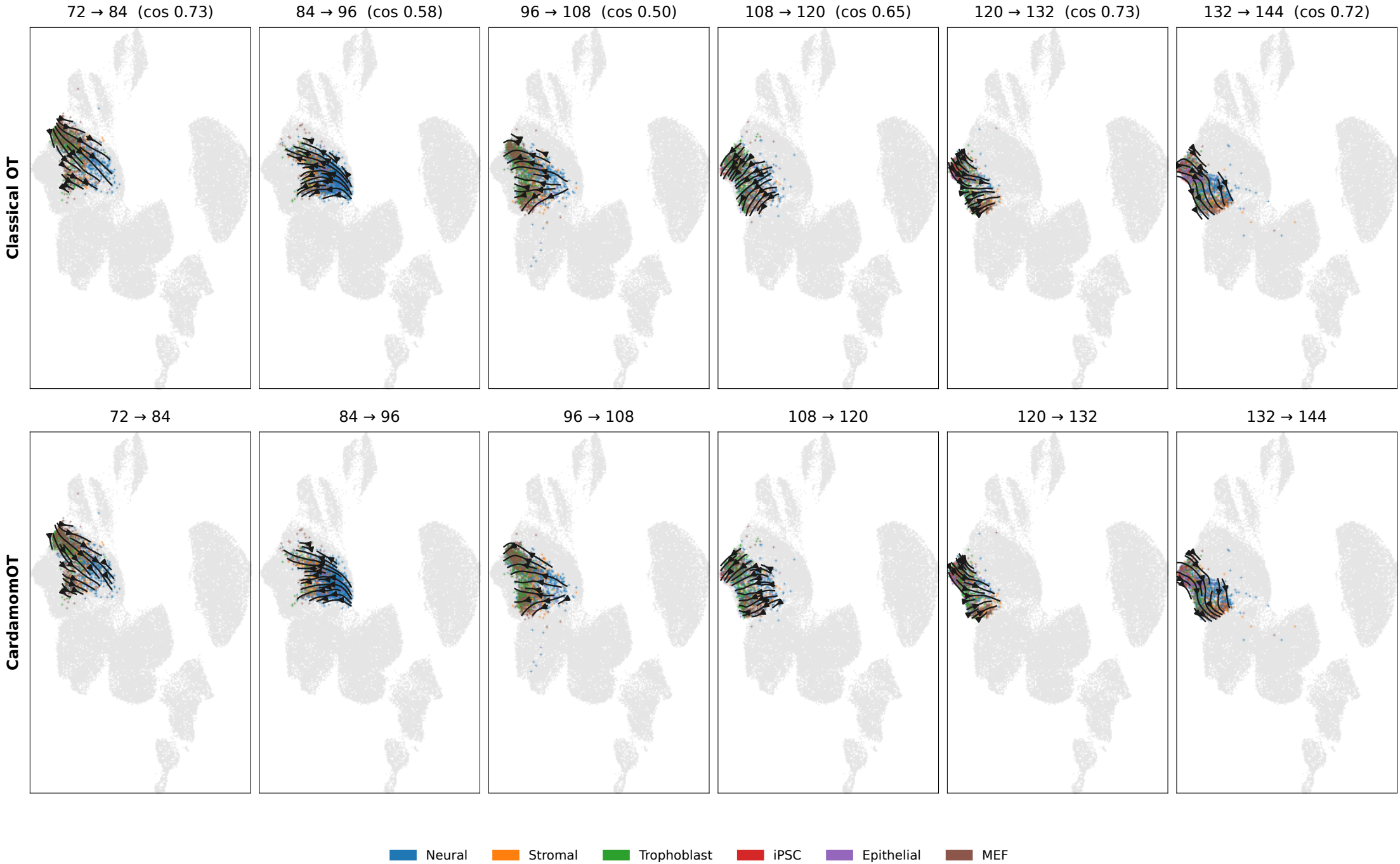
Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) towards their mean descendant at t_to, kNN projection among the cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



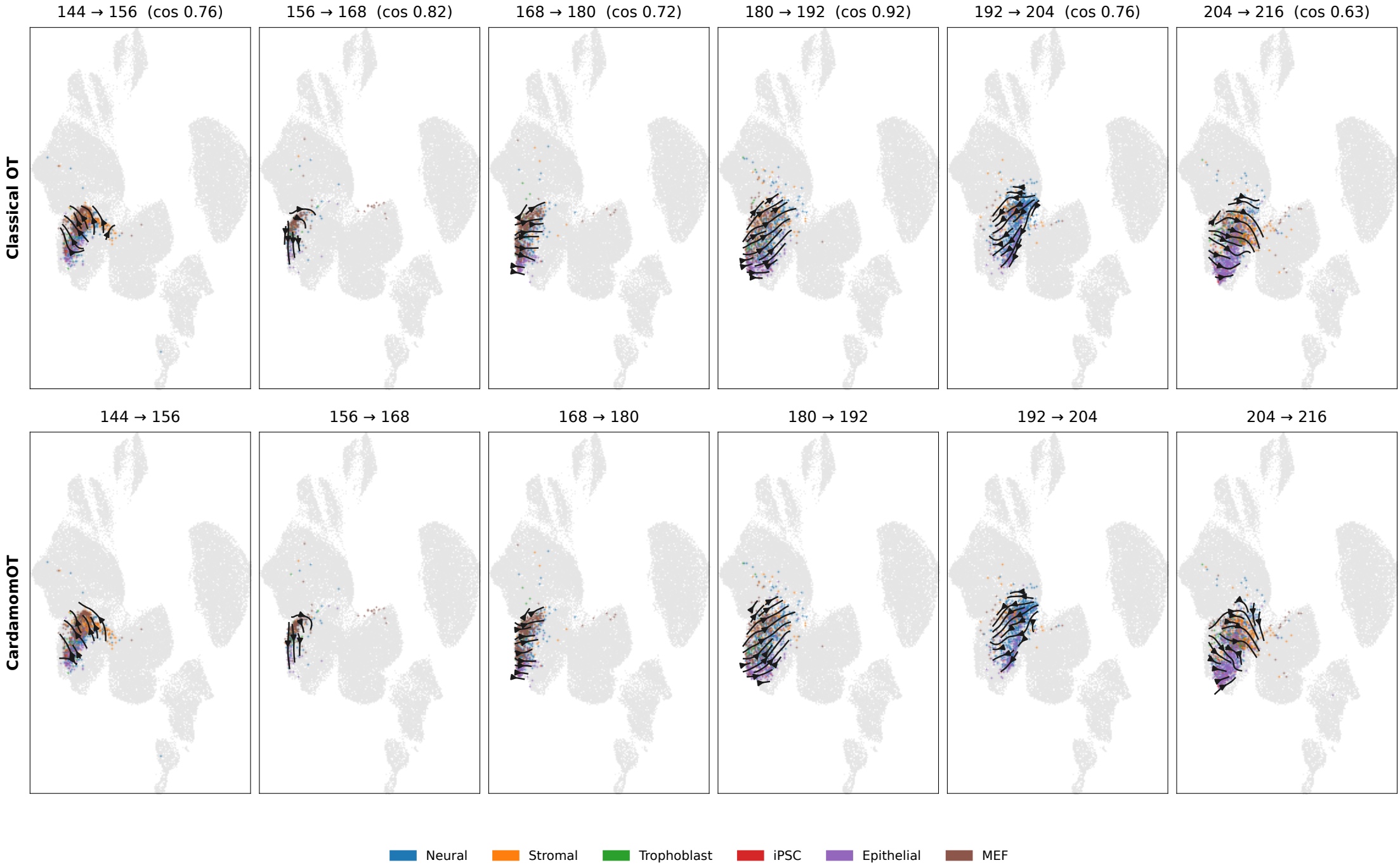
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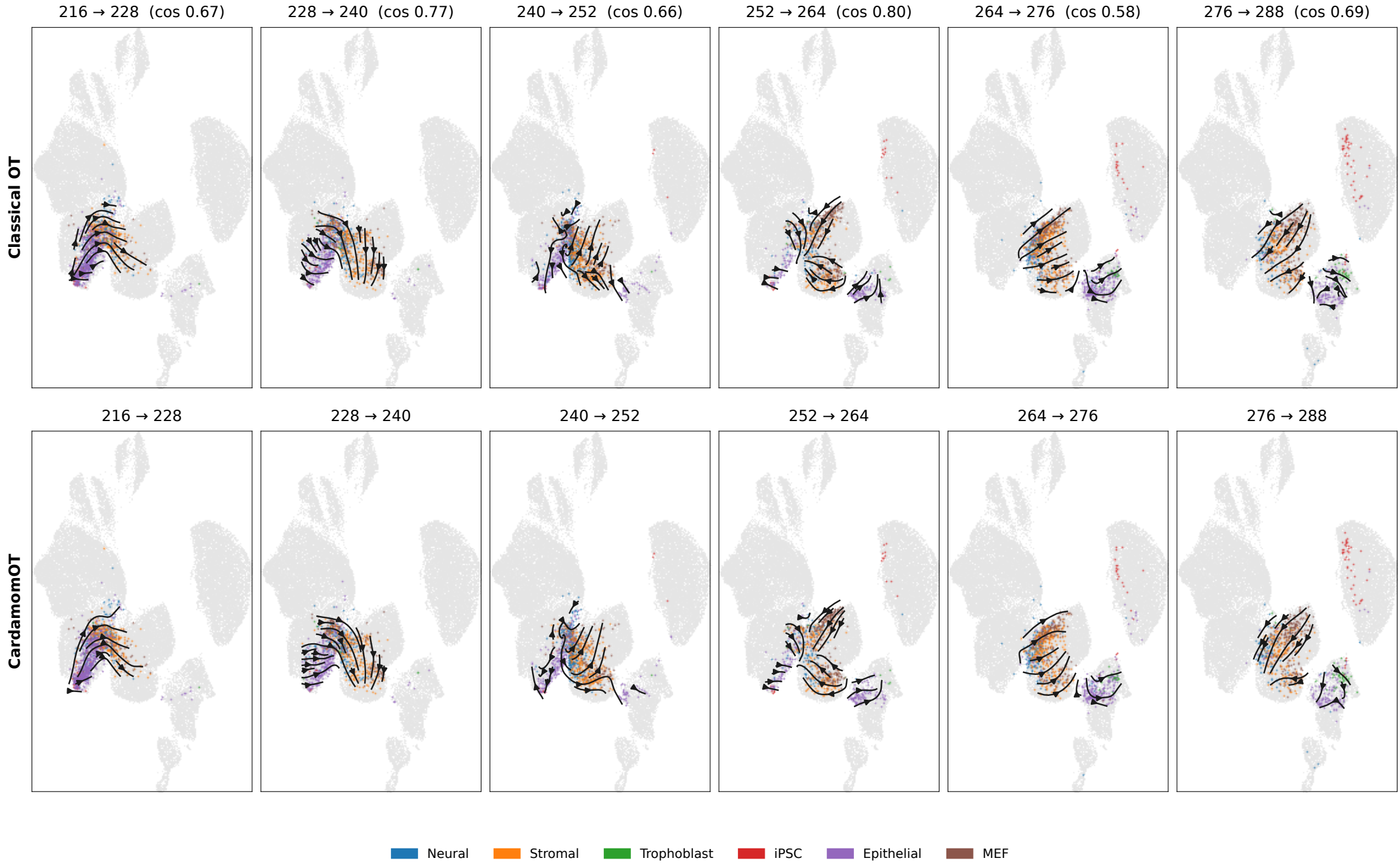
Teaser — displacement fields over time, classical OT vs CardamomOT

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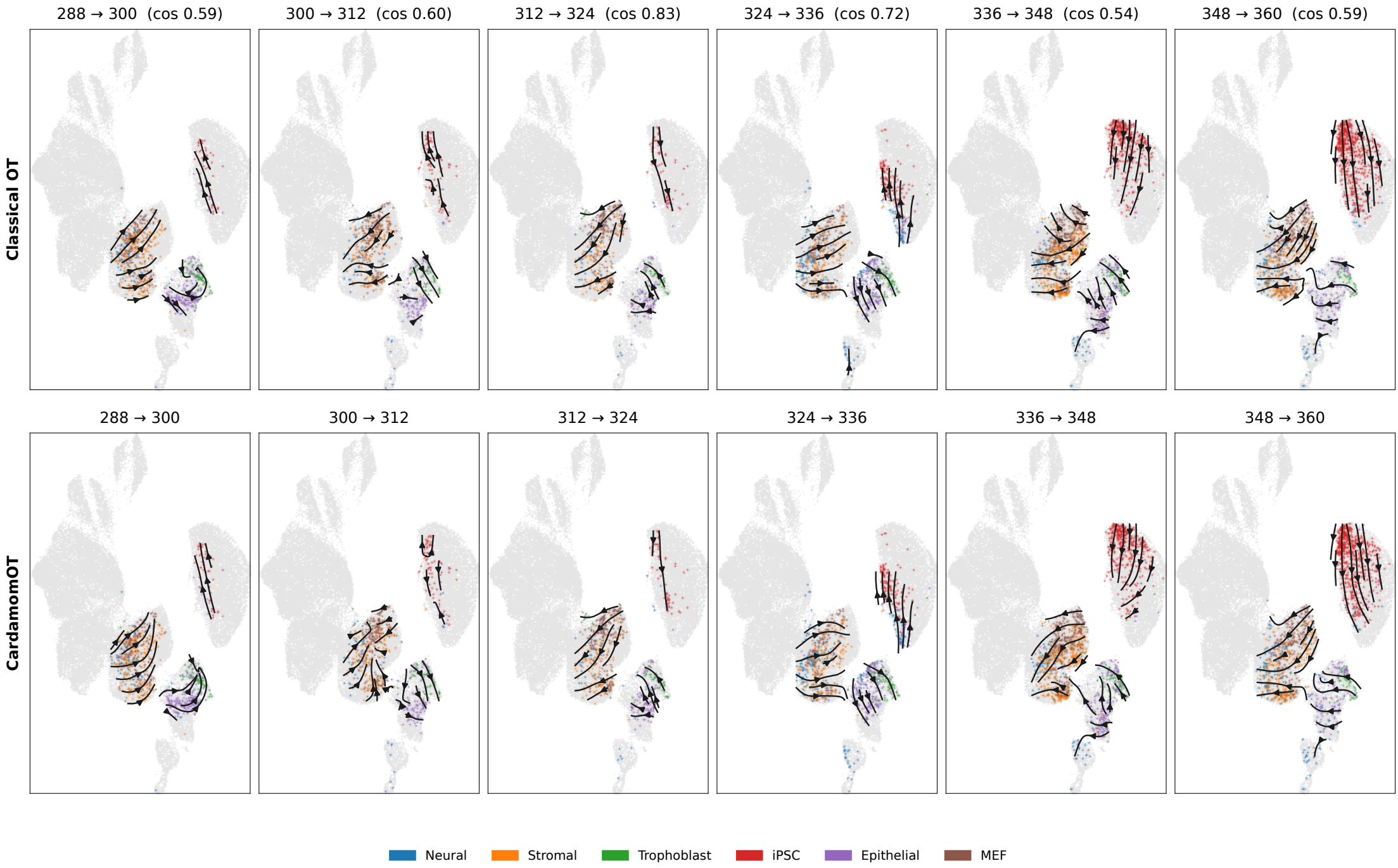
Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) towards their mean descendant at t_to, kNN projection among the cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



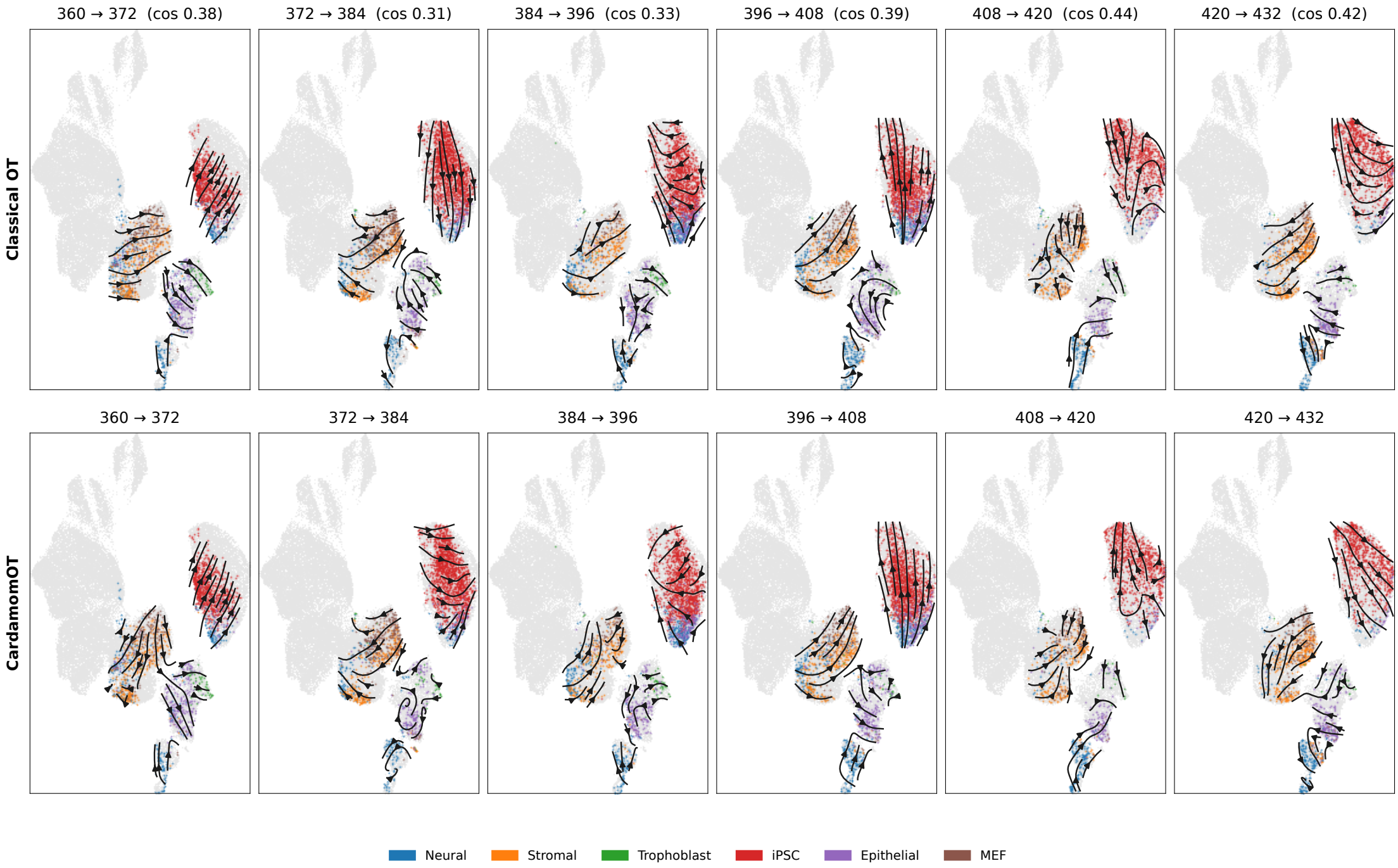
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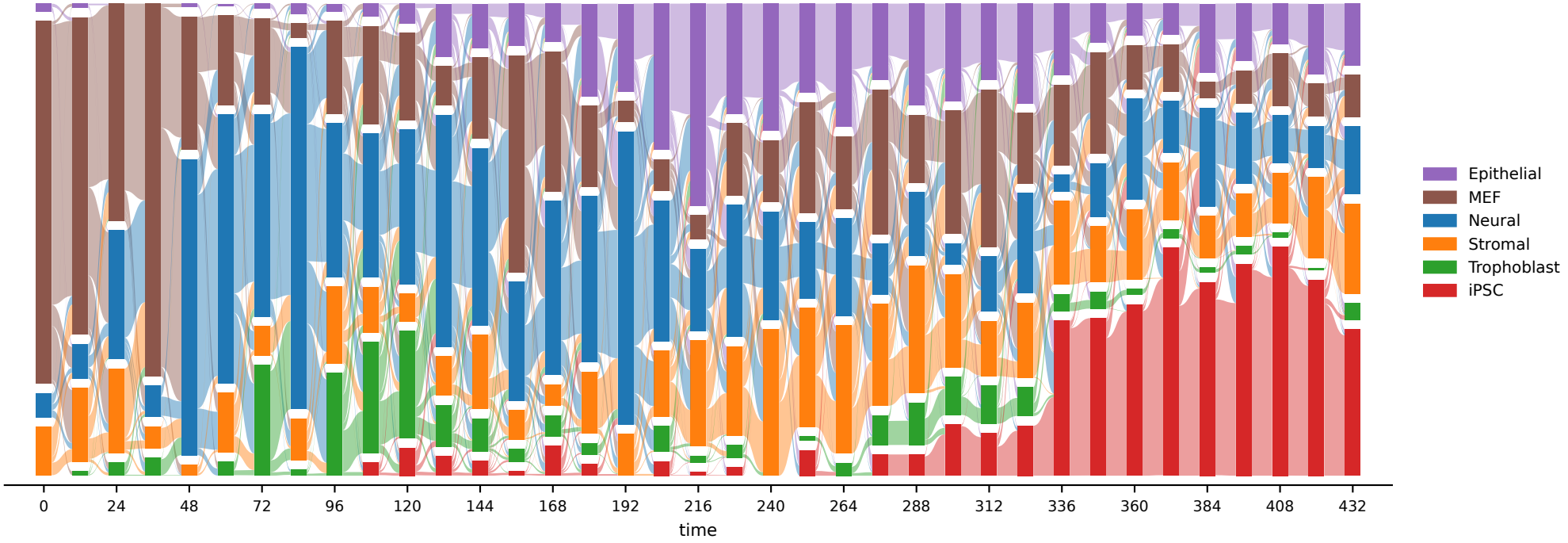
Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) towards their mean descendant at t_to, kNN projection among the cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



Classical OT — cell-type transitions and fate genes

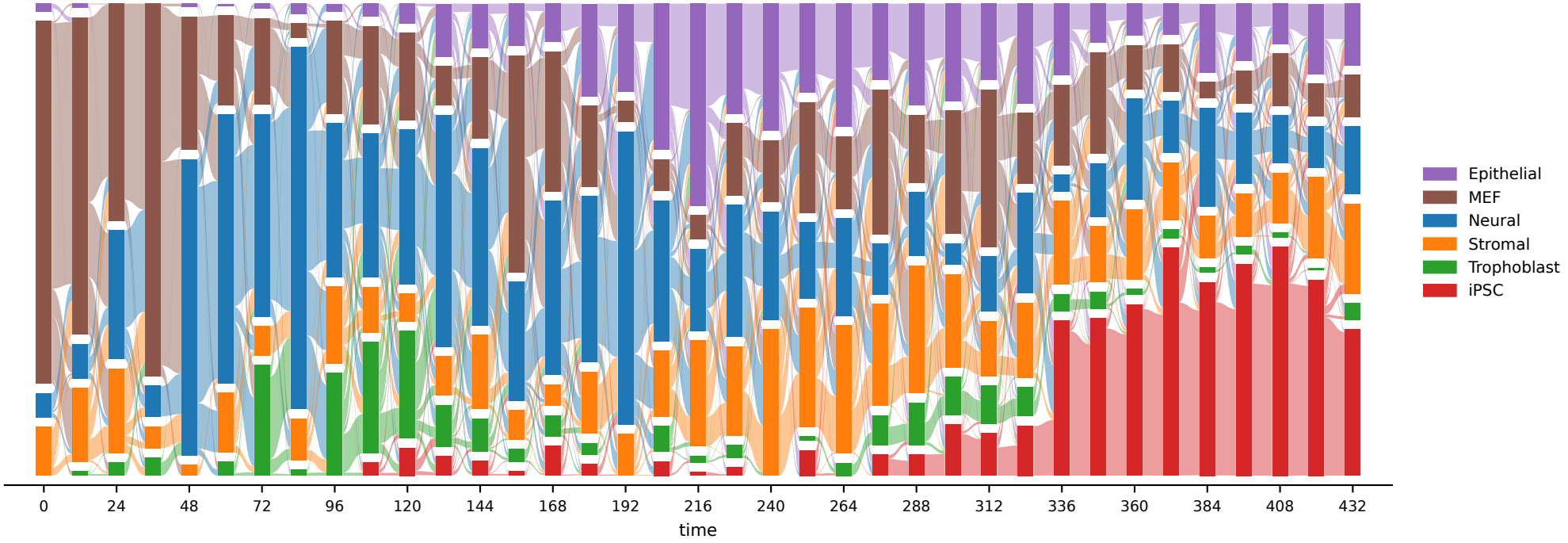
Waddington-OT couplings (every gene, train cells, per sample, growth from the net rate), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (classical_OT/fate_genes.csv).



	Top-10 genes up in the ancestors (vs ancestors of the other cells)	Top-10 genes up in the cell type (vs the other cells)
Epithelial	**Krt8** **Bex1** **Krt18** Bex4 **Spint2** **Cldn6** **Epcam** **Cldn7** **Krt19** **Cdh1**	**Krt8** **Krt18** **Epcam** **Cldn7** **Krt19** **Cldn6** **Cdh1** **Spint2** **Bex1** Dsp
MEF	**Col1a2** **Cald1** Timp1 **Thbs1** **Tagln** Col1a1 Lox Vim Timp2 **S100a6**	**Cald1** **S100a6** **Col1a2** Lmna Fstl1 Ahnak Flna Itgb1 **Tagln** **Thbs1**
Neural	**Vim** Cd109 **Cd24a** **Ccnd2** **S100a6** Limd2 **Avpi1** **S100a10** **Hmgn3** **Id1**	**Vim** **Ccnd2** **Id1** **Cd24a** H2afz **S100a6** **Avpi1** Tmsb4x **S100a10** **Hmgn3**
Stromal	**Sparc** **Lgals1** **Mgp** **Vim** **Bgn** **Col1a2** Pcolce **Col1a1** Aebp1 **Serpinh1**	**Sparc** **Lgals1** **Vim** **Mgp** **Serpinh1** **Col1a2** **Bgn** Gabarap Emp3 **Col1a1**
Trophoblast	**Ranbp1** **Mrpl12** **Cycs** **Tagln2** H2afz **Lgals3** Hsp90aa1 **Sri** Cks1b Nhp2	Dbi **Cycs** **Sri** Gng12 Anxa2 **Lgals3** **Tagln2** **Ranbp1** **Mrpl12** S100a6
iPSC	**Dppa5a** **Hsp90aa1** **Chchd10** **Mkrn1** **Apoe** **Bex1** **Fkbp3** **Mt2** **Fabp3** **Slc2a3**	**Dppa5a** **Hsp90aa1** **Chchd10** **Mkrn1** **Bex1** **Apoe** **Fkbp3** **Mt2** **Fabp3** **Slc2a3**

Teaser — CardamomOT: cell-type transitions and fate genes

Final soft couplings of the CardamomOT trajectories (model genes, data_<split> cells), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (cardamomOT/teaser_fate_genes.csv).



	Top-10 genes up in the ancestors (vs ancestors of the other cells)	Top-10 genes up in the cell type (vs the other cells)
Epithelial	**Krt8** **Krt18** **Epcam** **Krt7** **Mbnl3** **S100a14** Dbi **Bbx** Hmgb2 Rbbp7	**Krt8** **Epcam** **Krt18** **Krt7** **Mbnl3** **S100a14** **Bbx** Marcksl1 Cdkn1c Hcfc1r1
MEF	**Col1a2** **Vim** **Emp3** Sparc **Thbs1** Col1a1 Timp2 **Tagln** **Tpm1** Timp3	**Vim** **Tpm1** **Col1a2** **Emp3** Cd44 **Tagln** **Thbs1** Ccnd2 Xist S100a10
Neural	**Vim** **Ccnd2** **Dmrtc2** **Myc** **Mras** **Pcolce2** **Cenpa** S100a10 **Cks1b** **Xist**	**Dmrtc2** **Vim** **Ccnd2** **Mras** Id3 **Myc** **Pcolce2** **Cenpa** **Xist** **Cks1b**
Stromal	**Mgp** **Sparc** **Vim** **Col1a2** **Col1a1** **Timp2** **Emp3** **Selm** Id3 **Cd81**	**Mgp** **Sparc** **Vim** **Col1a2** **Col1a1** **Emp3** Xist **Timp2** **Selm** **Cd81**
Trophoblast	**Cks1b** **Hmgb2** **Dbi** **Sri** **Hsp90aa1** Cenpa **Hmgn5** Krt8 Mbnl3 Krt7	**Dbi** **Sri** **Hsp90aa1** S100a10 **Cks1b** **Hmgb2** Rbbp7 **Hmgn5** Cyb5r3 Ebna1bp2
iPSC	**Dppa5a** **Hsp90aa1** **Mybl2** **Nanog** **Ooep** **Sox2** **Nfatc2ip** **Gjb3** **Zfp42** **Cdc5l**	**Dppa5a** **Hsp90aa1** **Mybl2** **Ooep** **Nanog** **Cdc5l** **Gjb3** **Nfatc2ip** **Zfp42** **Sox2**

Genes of the model (100)

Data/data_train.h5ad, alphabetical order, coloured by selection role. Italics: no edge inside the selection (probability ≥ selection_edge_prob, literature-feasible;
0 genes). Trajectory preservation (classical OT on these genes vs every gene; network otvelo_granger): velocity cosine 0.88 (random genes 0.77), fate JS distance …

Antxr1	Efemp2	Ltbr	S100a10
Anxa3	Ehd2	Marcksl1	S100a14
Batf3	Emp3	Mbnl3	Sall4
Bbx	Epcam	Mgp	Sat1
Bhlhe40	Eva1b	Mras	Sdc4
Btg2	Fbn1	Msmo1	Selm
Ccnb1	Fmr1nb	Mybl2	Sgol1
Ccnd2	Foxo4	Myc	Sms
Cd44	Gata2	Nanog	Sox2
Cd81	Gjb3	Nasp	Sparc
Cdc5l	Gnl3	Nfatc2ip	Sri
Cdkn1c	Gpx2	Nfic	Tagln
Cenpa	Gpx3	Nfix	Taldo1
Cited2	Hcfc1r1	Obox6	Tcl1
Cks1b	Hmgb2	Ooep	Tgif1
Col1a1	Hmgn5	Pcolce2	Thbs1
Col1a2	Hsp90aa1	Phlda2	Timp2
Crabp2	Id3	Prnp	Timp3
Cyb5r3	Ier3	Prrx1	Tmbim1
Dbi	Igfbp3	Pttg1ip	Tnc
Ddit3	Klf4	Rbbp7	Tpm1
Ddx21	Krt18	Rhoc	Tspo
Dmrtc2	Krt7	Rhox6	Vim
Dppa5a	Krt8	Rhox9	Xist
Ebna1bp2	Laptm4a	Rrp9	Zfp42

perturbed (2) query (26) driver (34) entropy (38)

0. Per-cell sequencing depth

Diagnostic on the whole transcriptome within homogeneous groups (sample, time, cell type); counts are modelled as NB(k, c / s_i) when the depth factor s_i is applied.

Depth

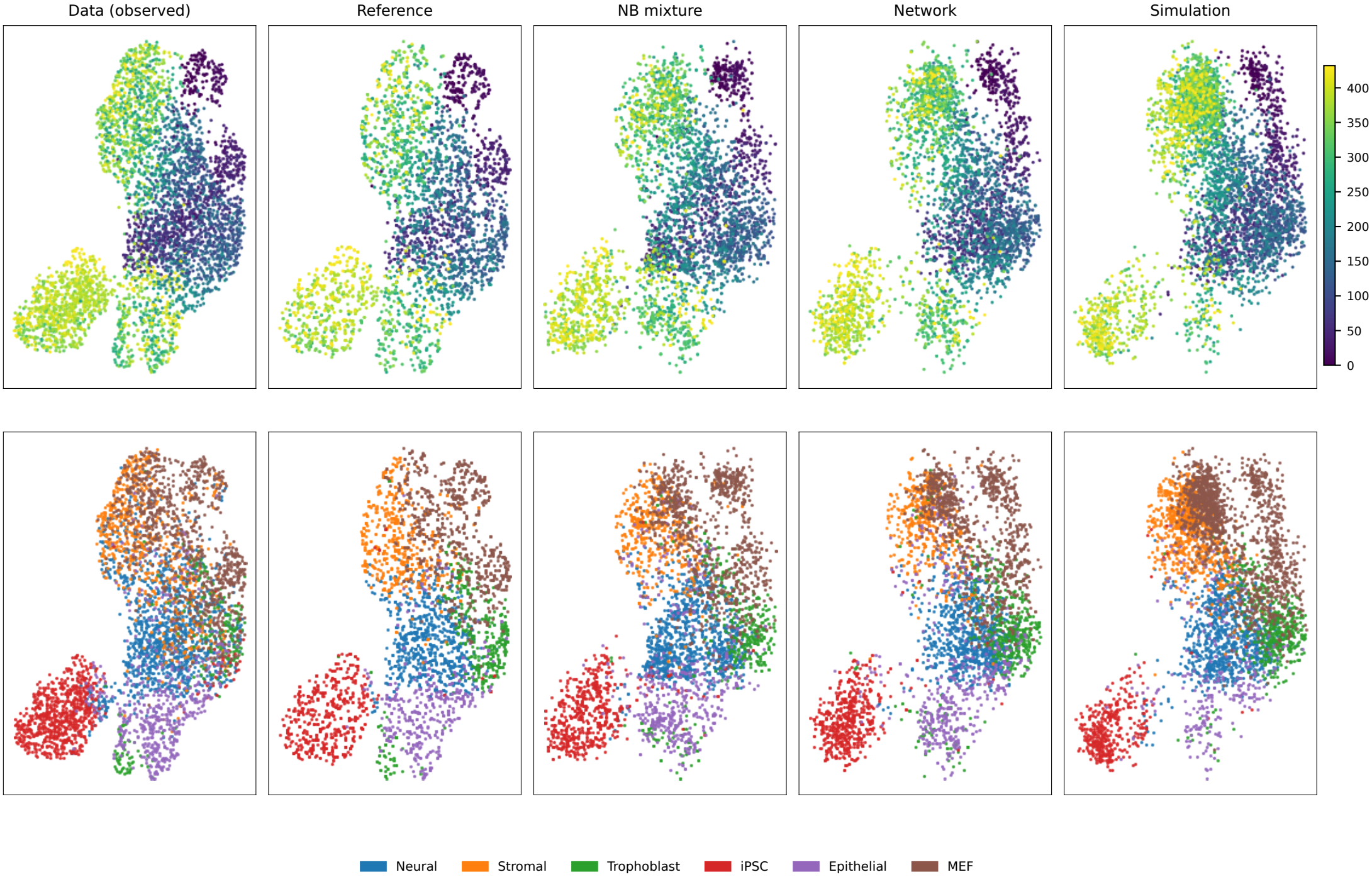
not estimated

Reason

4329 genes < 10000

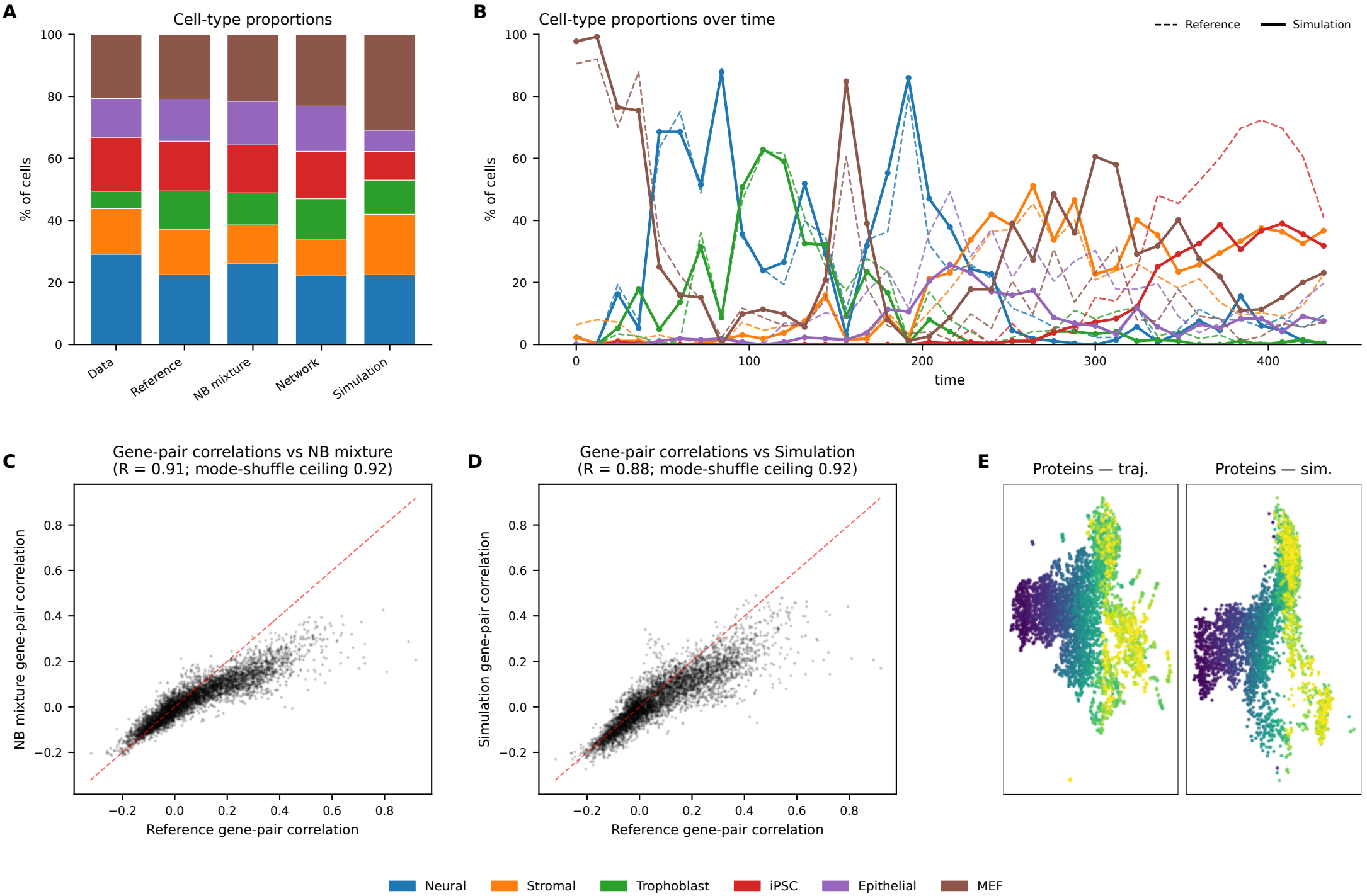
1. Generative model — trajectories and simulation

UMAP learned on the observed data; the trajectories (Reference: one state per ancestor, i.e. growth-weighted by exp ∫R_opt, as the simulation has proliferation), the NB mixture and network-driven modes along them and the full simulation are projected onto it.



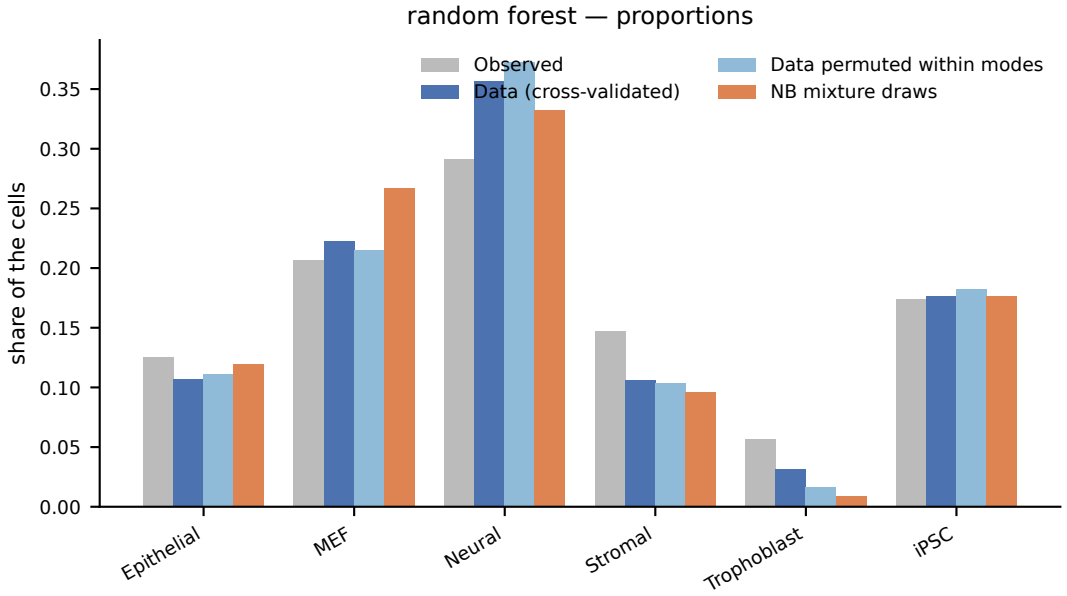
1. Generative model — quantitative checks

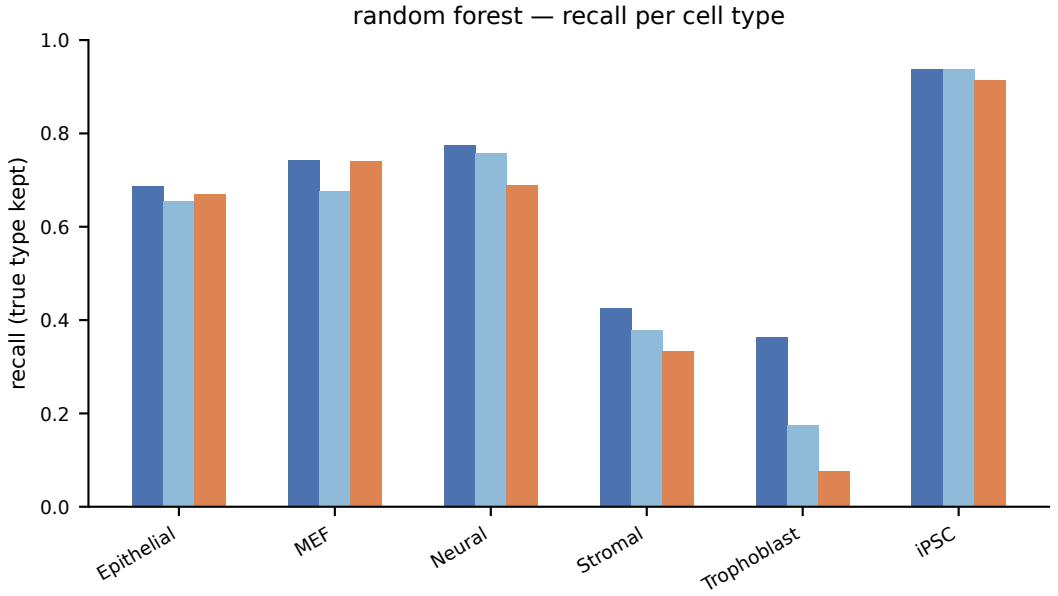
Reference = RNA of the trajectories equivalent to the simulation (growth-weighted: simulation with proliferation).

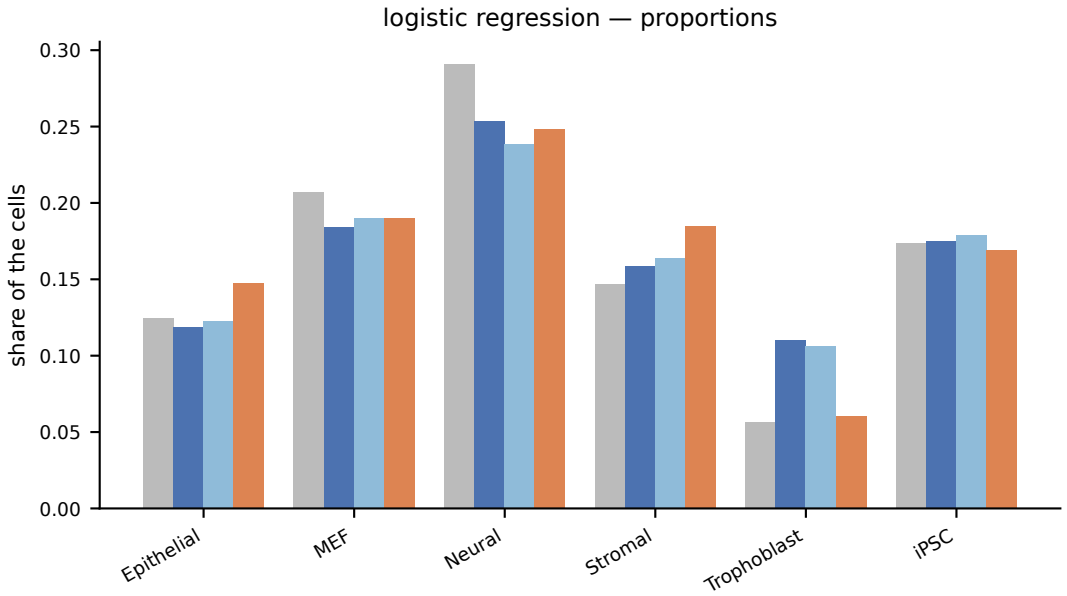


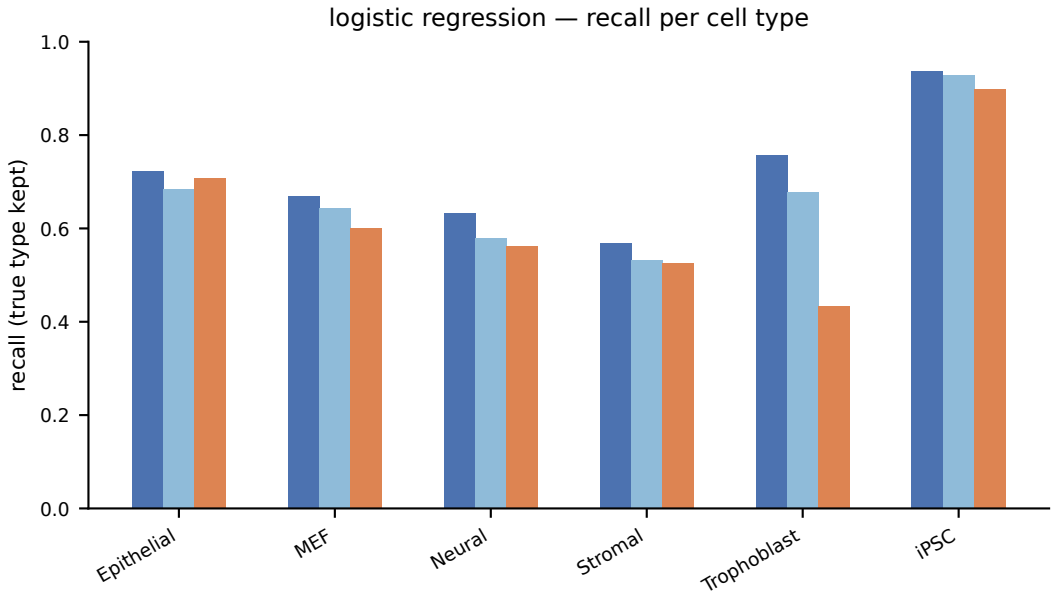
1. Generative model — cell-type identity in the NB mixture

One draw per real cell, per-sample classifiers trained on the other cells. Cross-validated data = classifier limit; permuted within modes (same sample, time, mode) = best model with these modes. Gap data → permuted: modes too coarse; permuted → draws: NB fit.







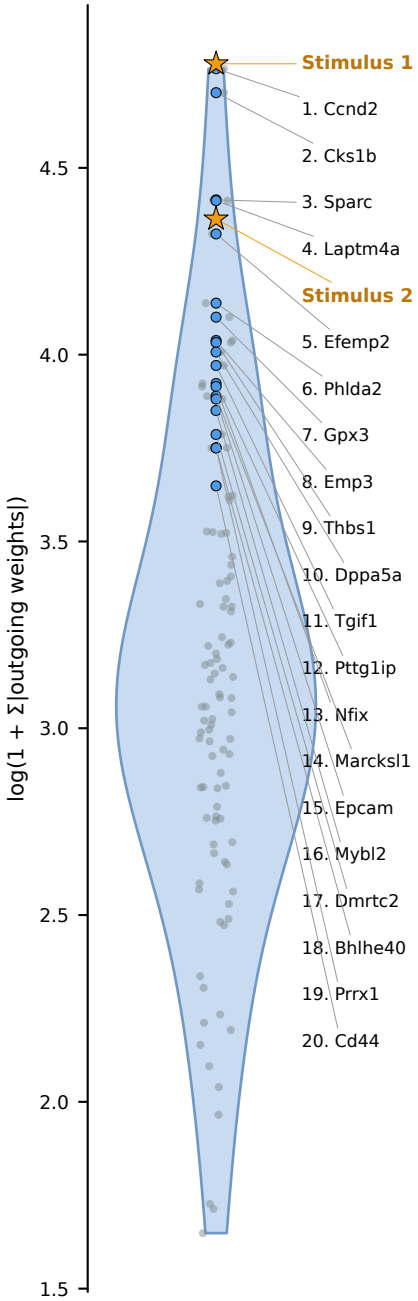


2. Gene regulatory network — regulatory power

Self-interactions excluded. Top 20 genes labelled; stimulus shown as a star (stimulus = 1.0 ≥ 0.5).

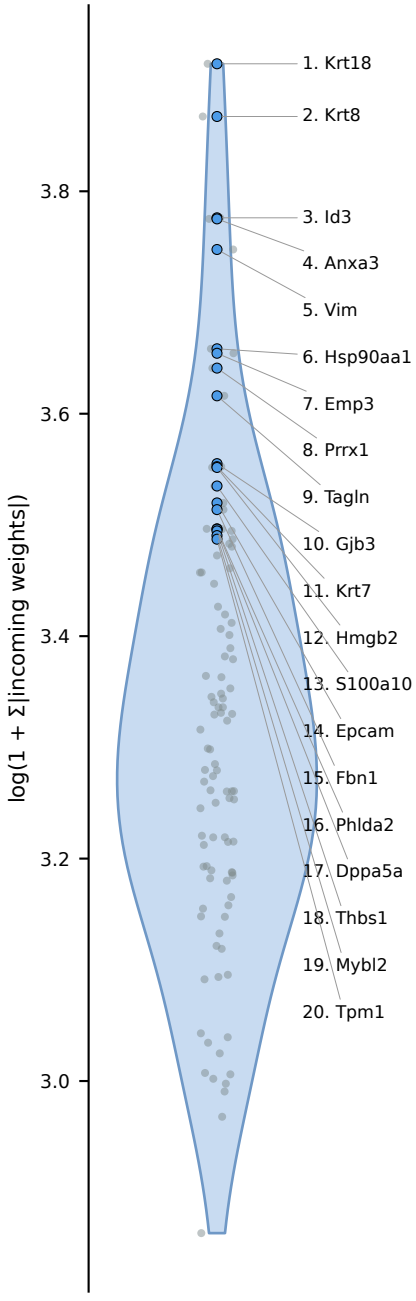
A

Outgoing regulation (regulators)



B

Incoming regulation (targets)



C Top 20 regulators

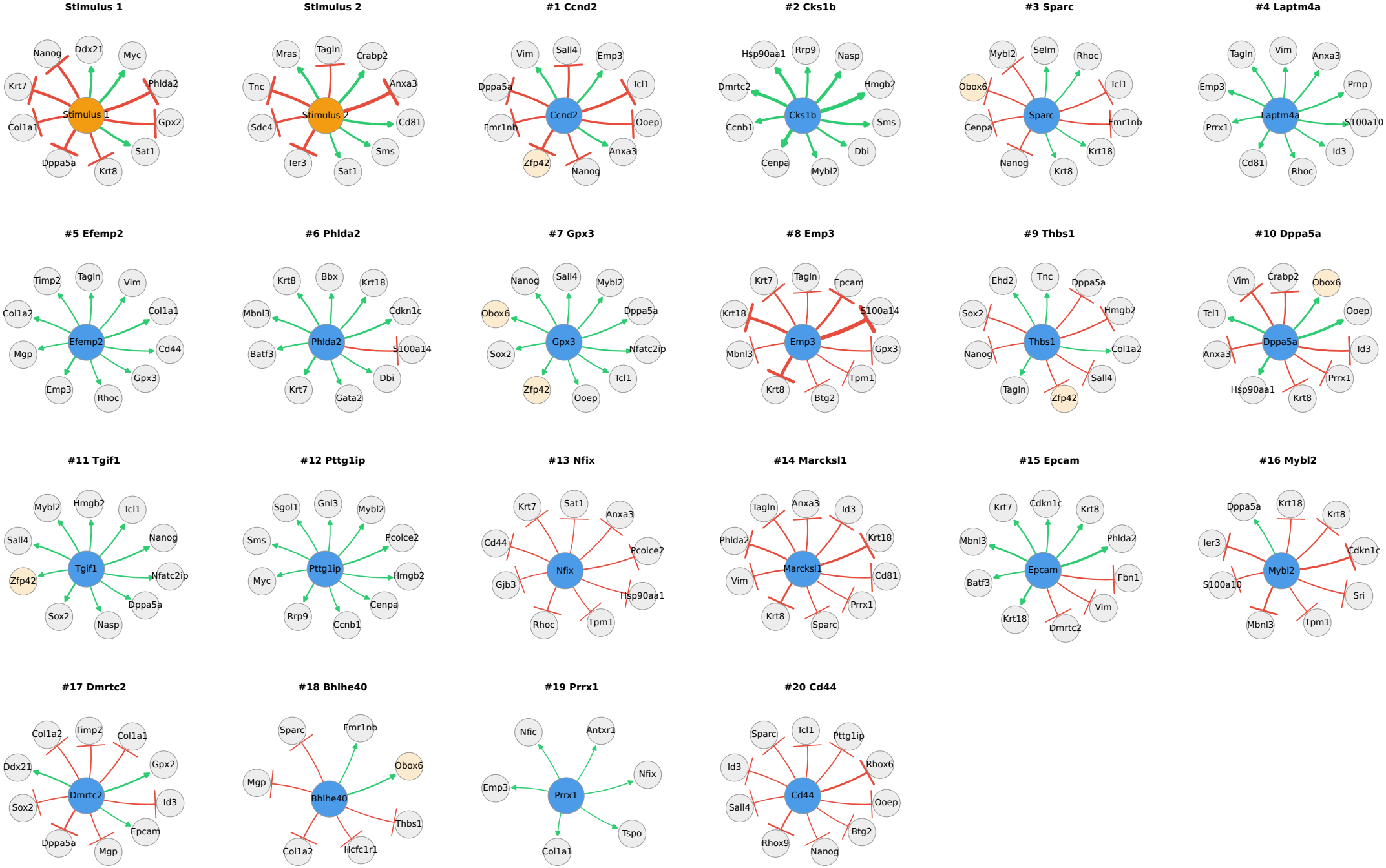
#	Regulator	Σ|w|	act.	inh.
1	Ccnd2	116.38	66	27
2	Cks1b	109.09	48	39
3	Sparc	81.62	62	31
4	Laptm4a	81.45	72	17
5	Efemp2	74.43	72	11
6	Phlda2	61.67	40	34
7	Gpx3	59.36	35	51
8	Emp3	55.71	9	73
9	Thbs1	55.41	57	25
10	Dppa5a	53.98	31	36
11	Tgif1	52.04	62	18
12	Pttg1ip	49.54	62	13
13	Nfix	49.14	4	84
14	Marcksl1	47.85	9	72
15	Epcam	47.48	31	37
16	Mybl2	46.01	24	58
17	Dmrtc2	43.09	33	42
18	Bhlhe40	41.55	25	55
19	Prrx1	41.52	81	9
20	Cd44	37.42	16	66
★	Stimulus 1	117.93	38	54
★	Stimulus 2	77.50	51	22

D Top 20 regulated genes

#	Target	Σ|w|	act.	inh.
1	Krt18	49.13	26	45
2	Krt8	46.81	36	38
3	Id3	42.65	28	46
4	Anxa3	42.60	32	42
5	Vim	41.42	28	51
6	Hsp90aa1	37.80	42	30
7	Emp3	37.64	25	46
8	Prrx1	37.13	18	65
9	Tagln	36.19	24	40
10	Gjb3	33.99	42	40
11	Krt7	33.90	29	39
12	Hmgb2	33.87	31	34
13	S100a10	33.29	33	43
14	Epcam	32.78	30	41
15	Fbn1	32.57	21	58
16	Phlda2	32.00	26	37
17	Dppa5a	31.99	19	28
18	Thbs1	31.94	24	58
19	Mybl2	31.80	31	35
20	Tpm1	31.69	33	41

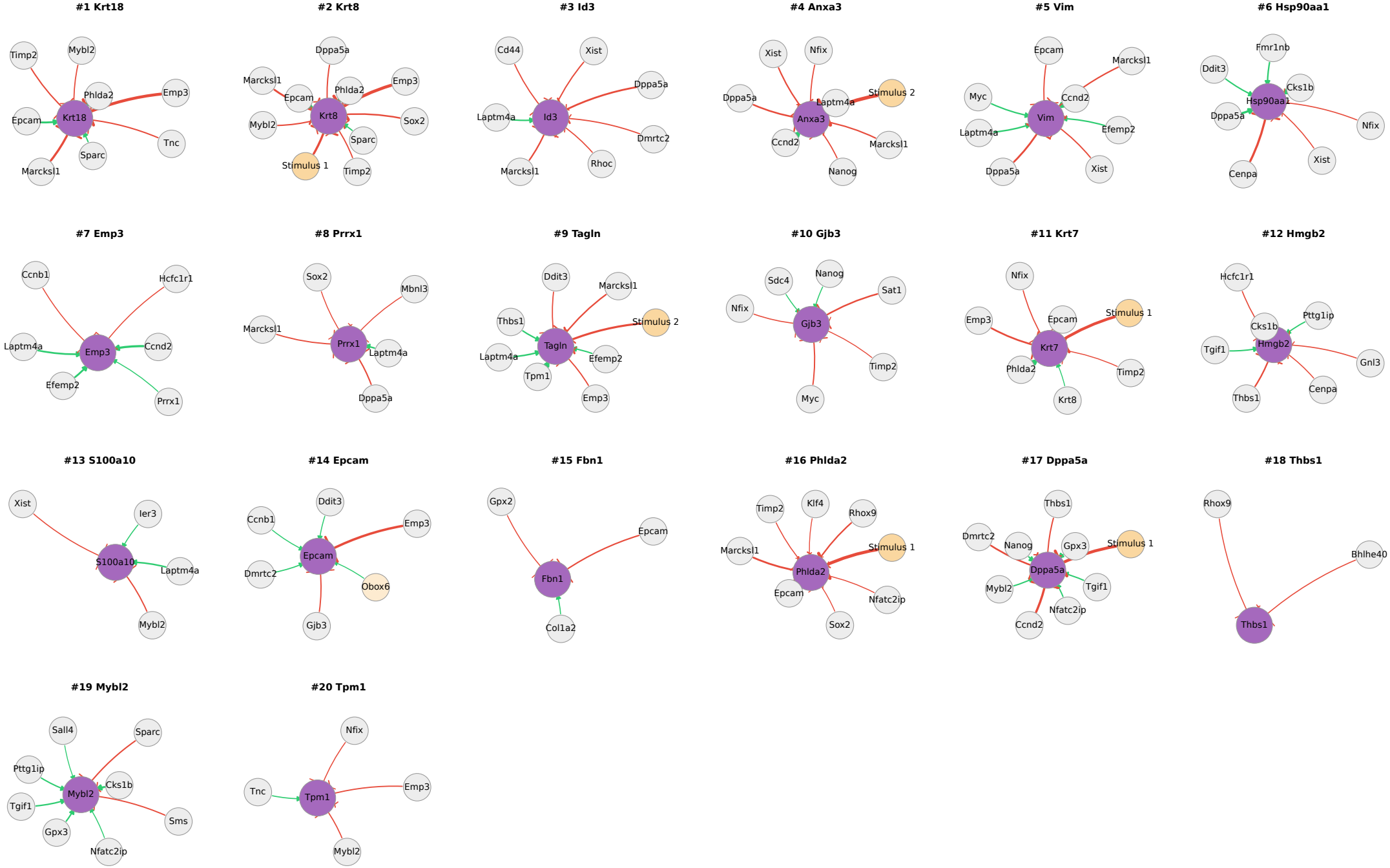
2. Gene regulatory network — top regulators and their main targets

Edges shown only if among the 10 strongest (|w|) targets of the regulator AND the 10 strongest regulators of the target; edge width ∝ |weight| (global scale).
Perturbed genes (KO/OV) are highlighted in beige.



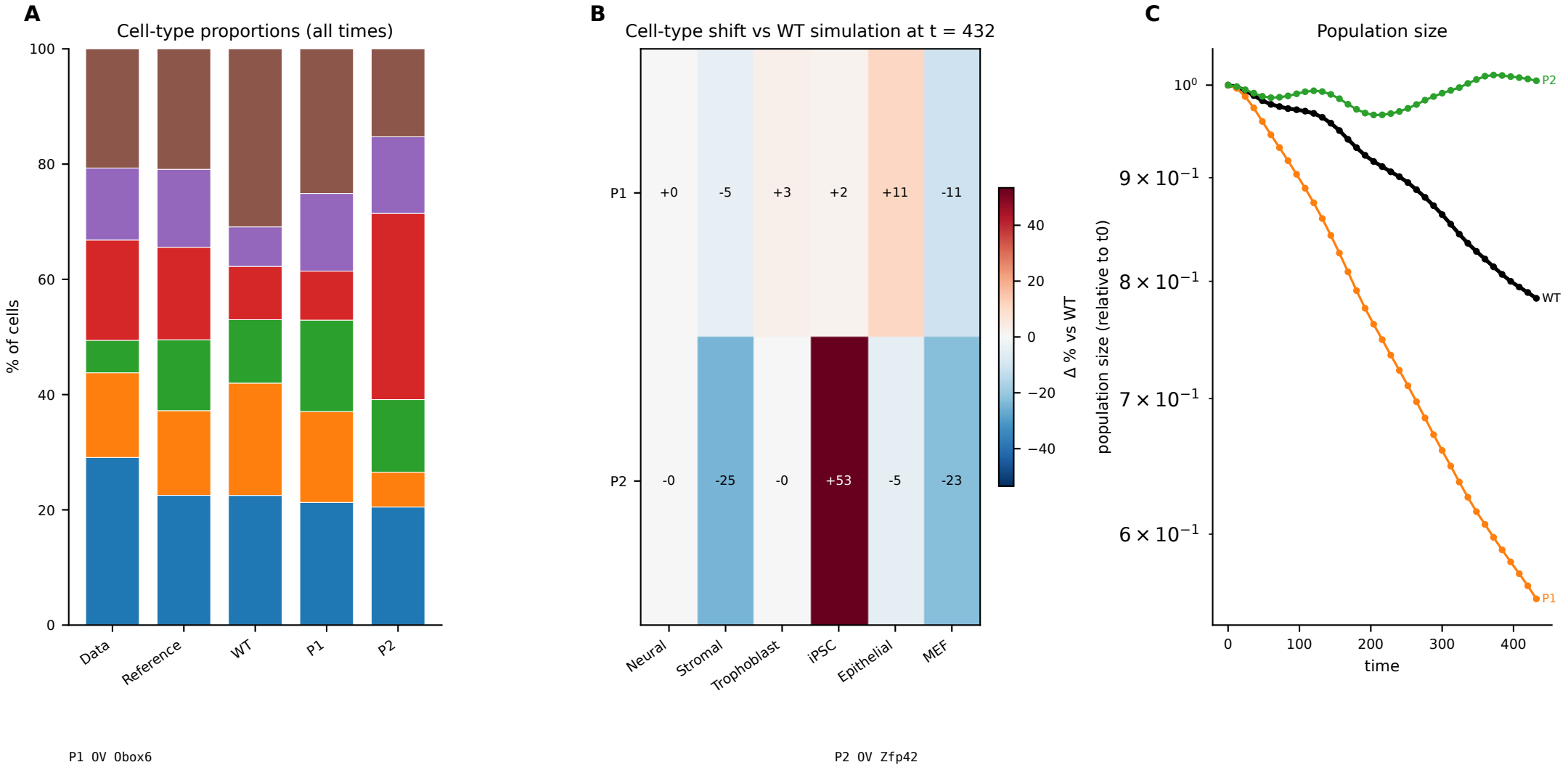
2. Gene regulatory network — top regulated genes and their main regulators

Edges shown only if among the 10 strongest (|w|) targets of the regulator AND the 10 strongest regulators of the target; edge width ∝ |weight| (global scale).
Perturbed genes (KO/OV) are highlighted in beige.



3. In-silico perturbations — overview

Cell types predicted by per-sample classifiers (logistic) trained on the observed data. Population sizes from the branching simulation (proliferation MLP + RATE effects).

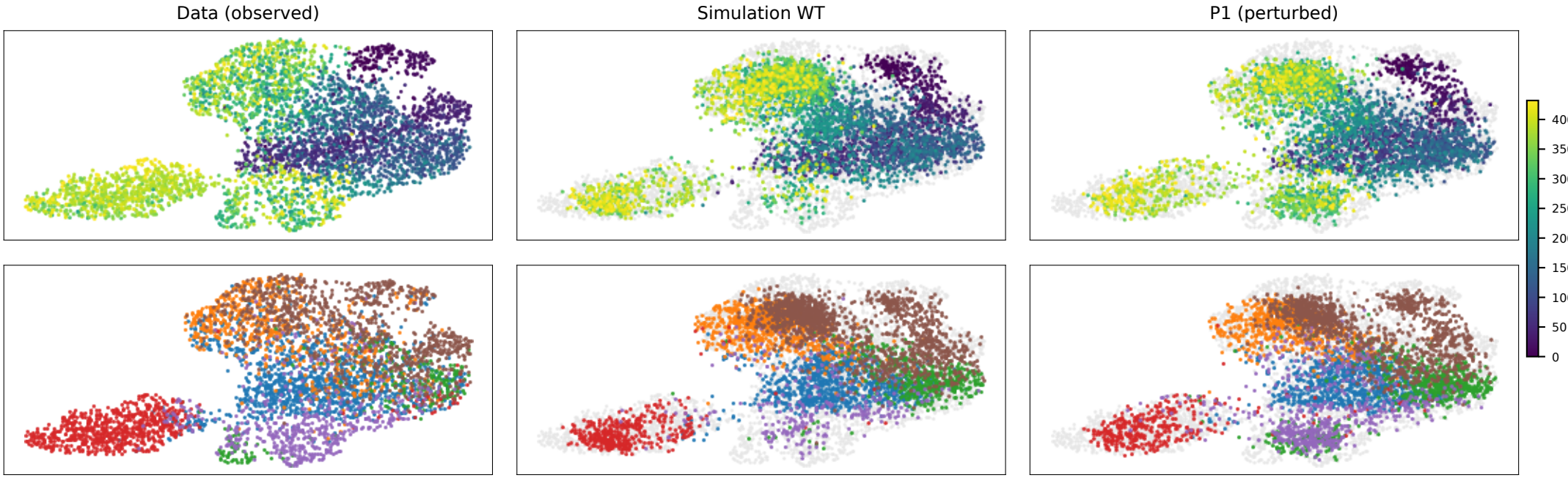


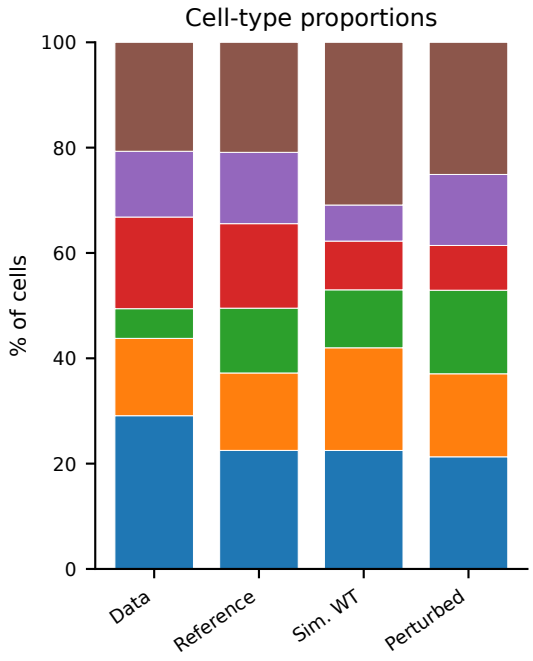
P1 OV Obox6

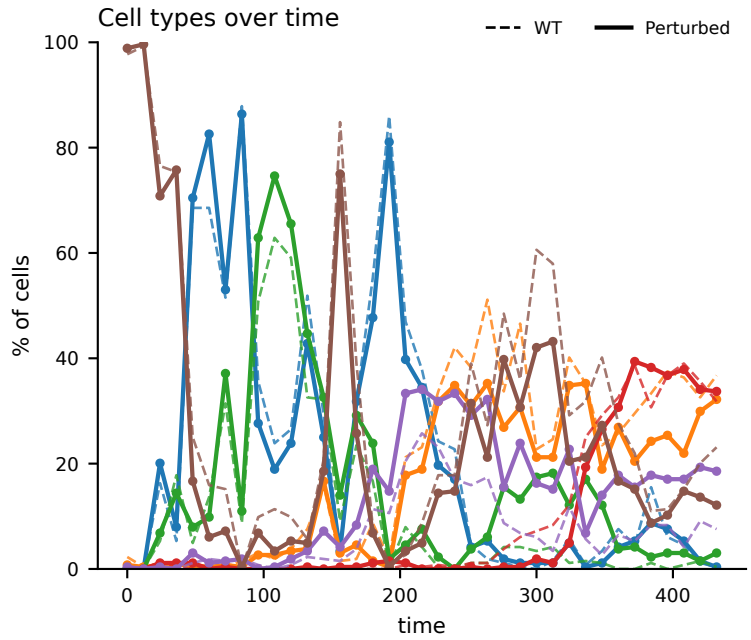
P2 OV Zfp42

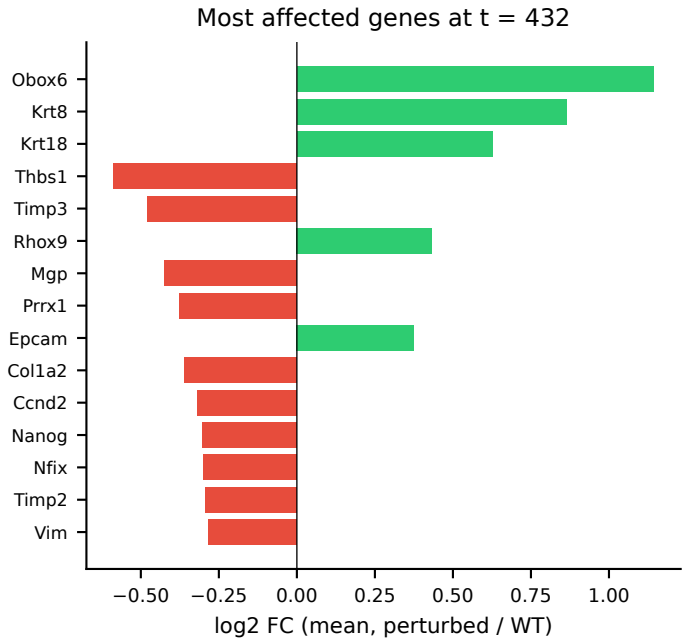
3. Perturbation P1 — OV Obox6

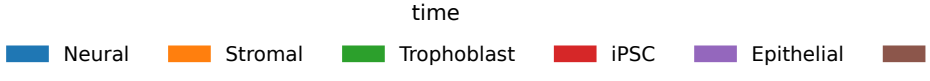
KO_none_OV_Obox6. UMAP: perturbed simulation projected on the WT embedding (grey = observed data). Population at t = 432: ×0.71 vs WT.





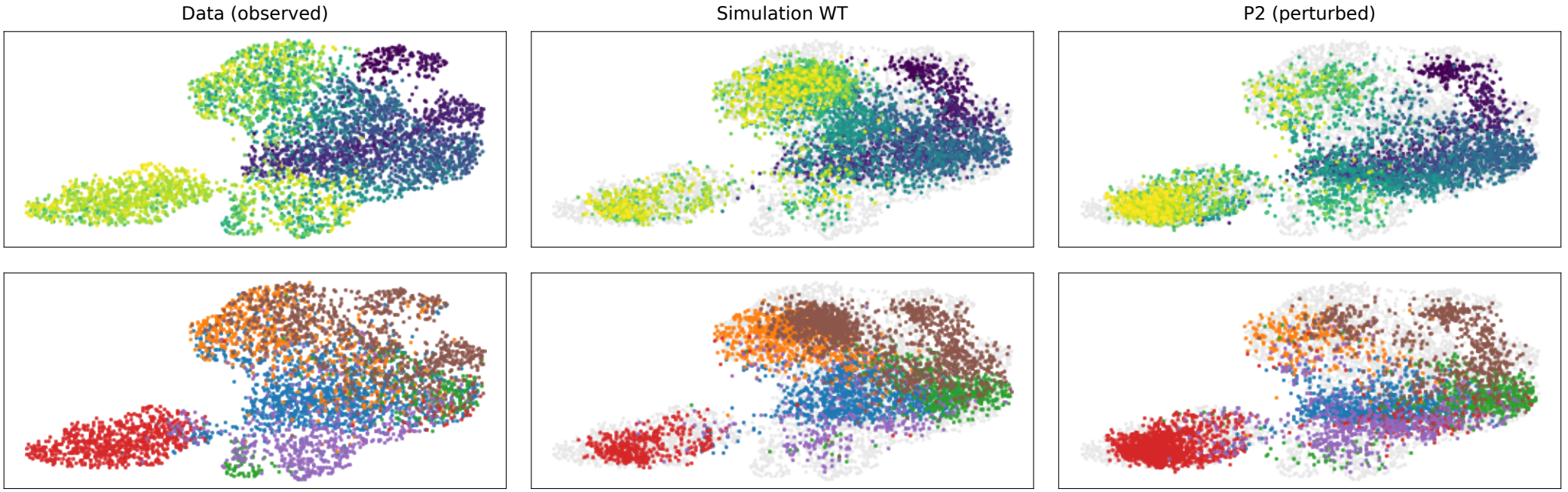


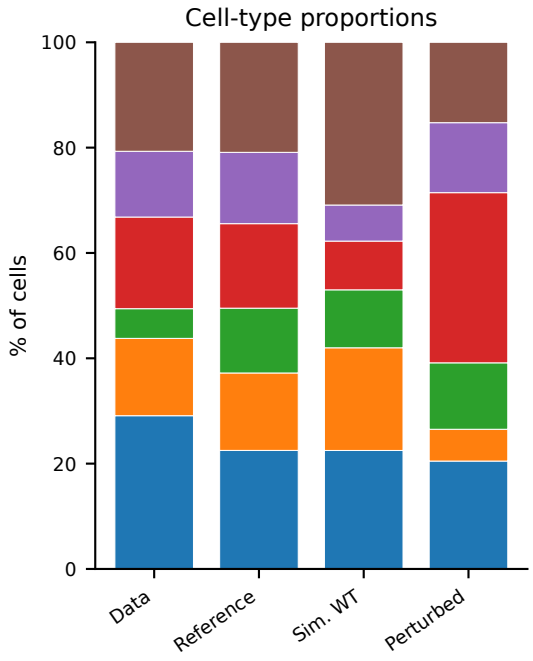


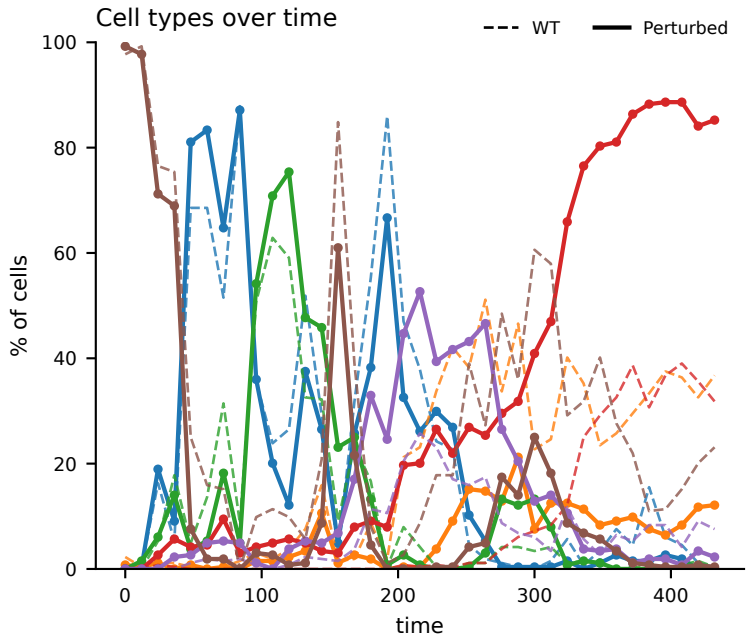


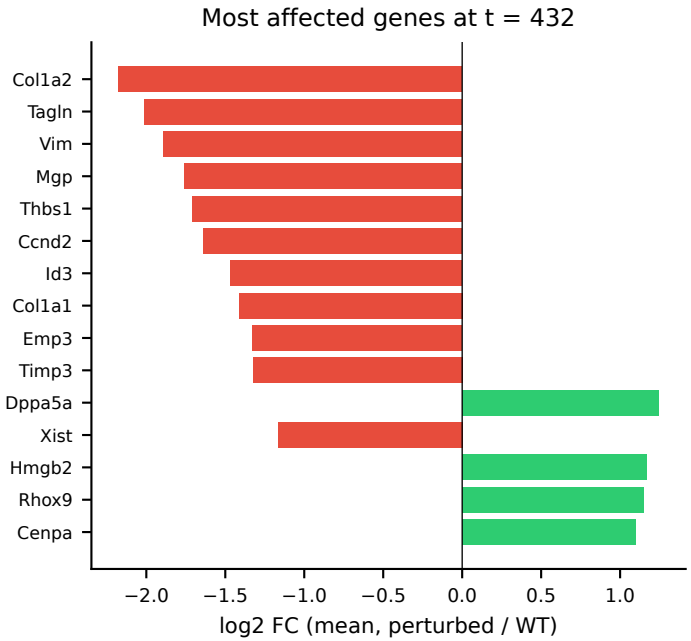
3. Perturbation P2 — OV Zfp42

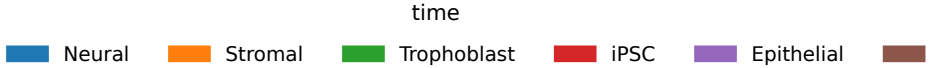
KO_none_OV_Zfp42. UMAP: perturbed simulation projected on the WT embedding (grey = observed data). Population at t = 432: ×1.28 vs WT.









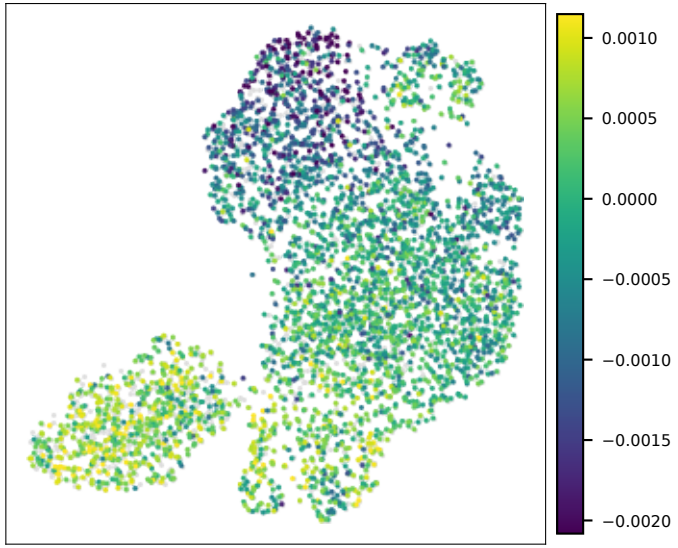


4. Proliferation — prior vs learned net rate (birth − death)

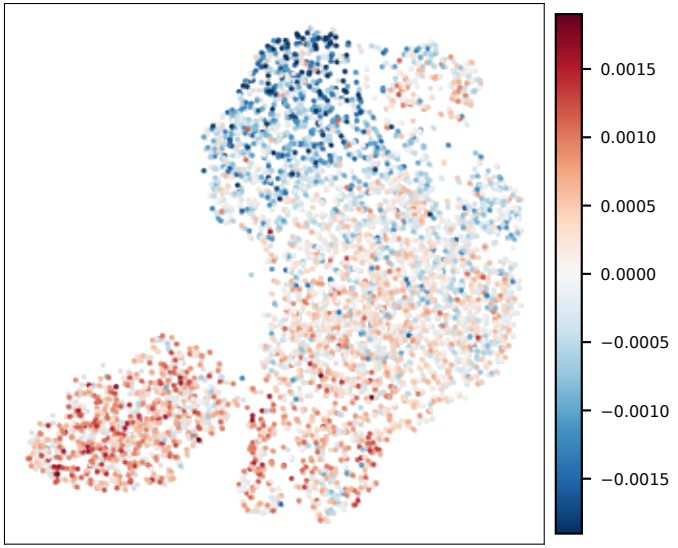
Learned = MLP R(P), mean over the trajectory states behind each observed cell (grey = cell not reached). Rates per time unit of the data.

No prior provided:
uniform net rate
(only relative rates are learned)

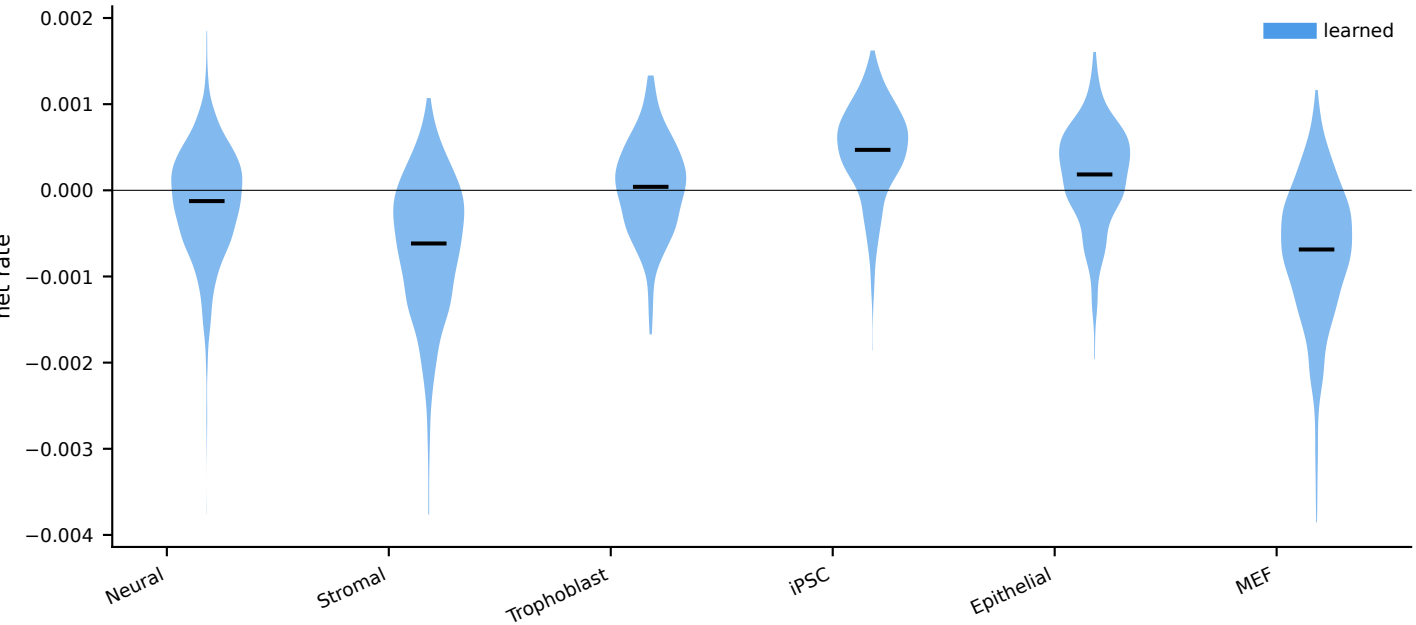
Learned net rate (MLP R(P))



Learned − mean



Net proliferation rate per cell type (cell_type)

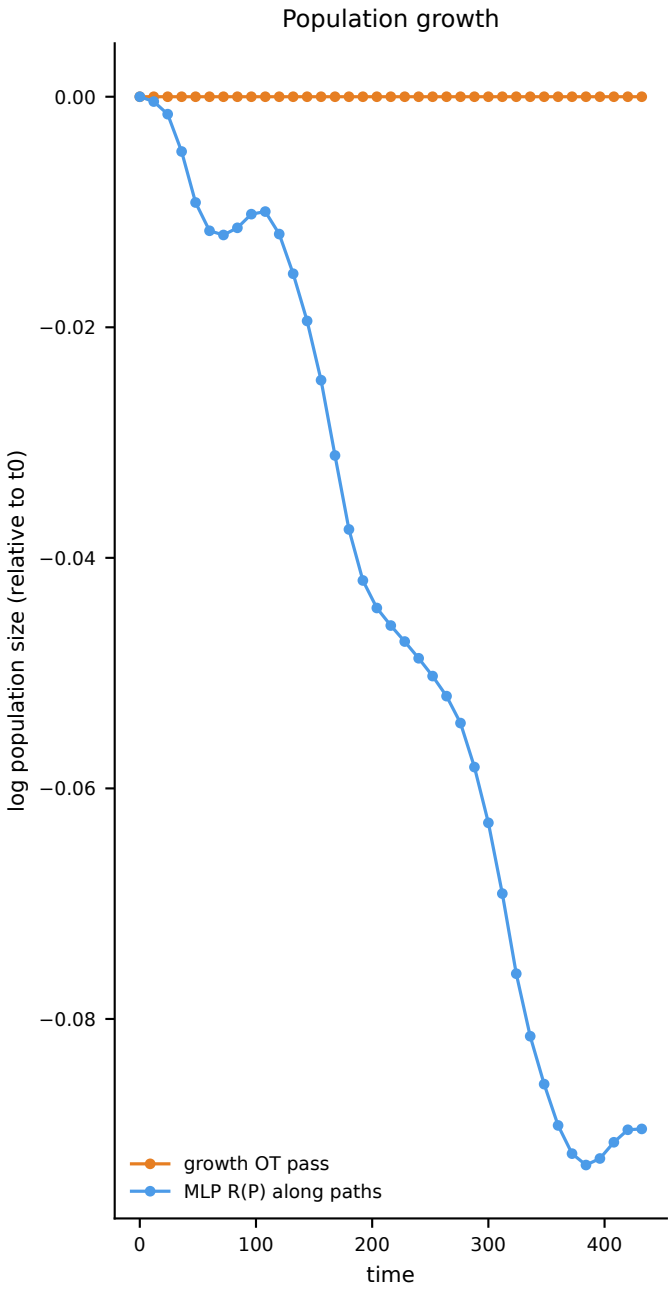


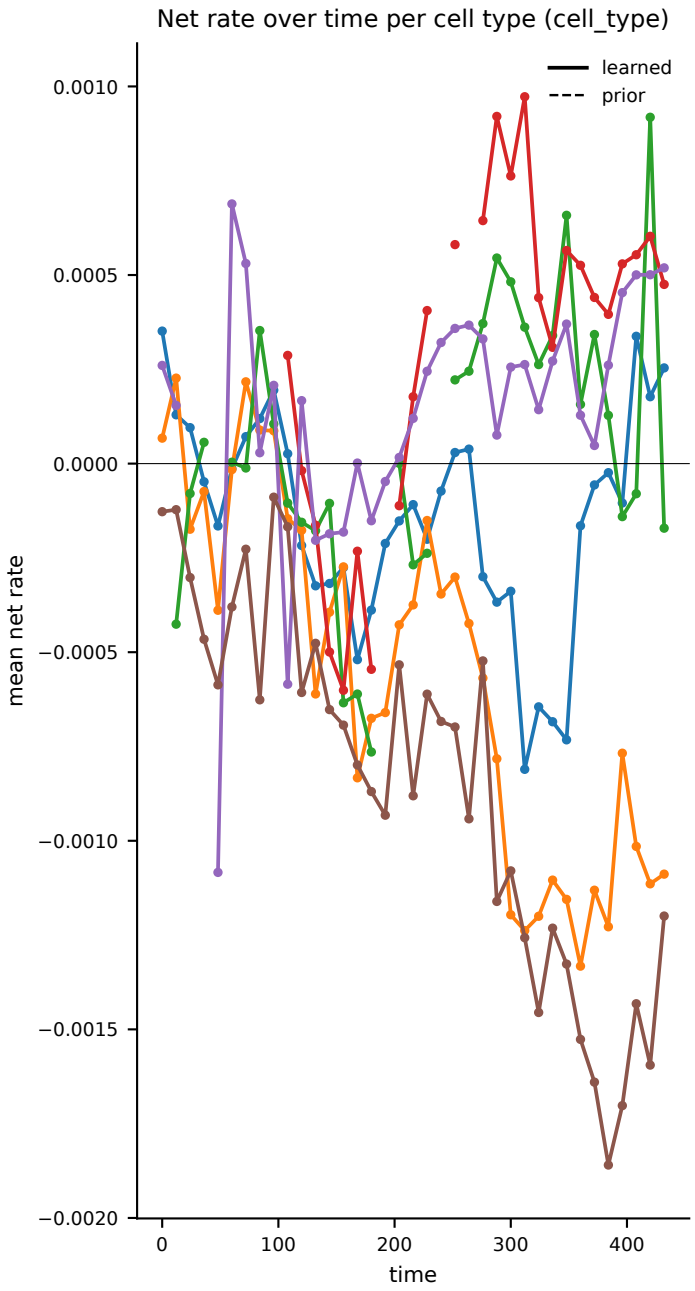
Mean net rate per cell type (cell_type)

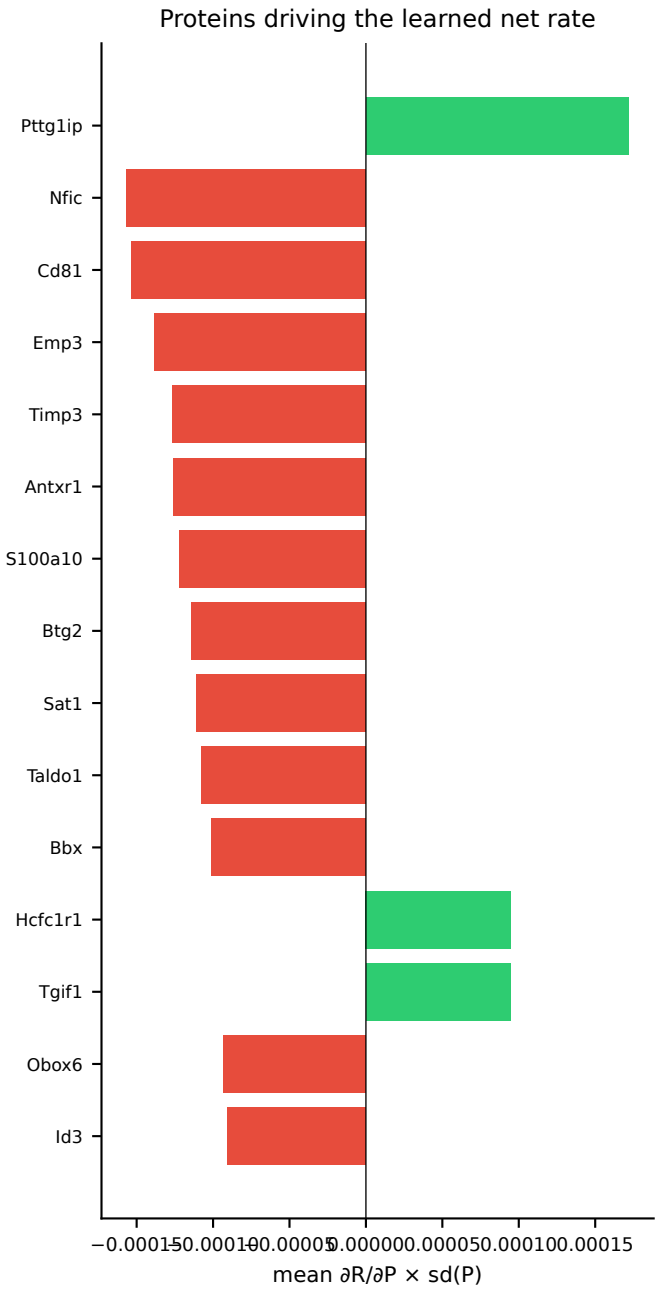
Cell type	Anchor	Prior	Learned
Neural	—	—	-0.000125
Stromal	—	—	-0.000617
Trophoblast	—	—	4.05e-05
iPSC	—	—	0.00047
Epithelial	—	—	0.000184
MEF	—	—	-0.000687

4. Proliferation — population growth and drivers

Relative population size from the mean growth factor per interval; its absolute level follows the prior (anchored on the sheet population_sizes by fit_population_anchors, if filled).



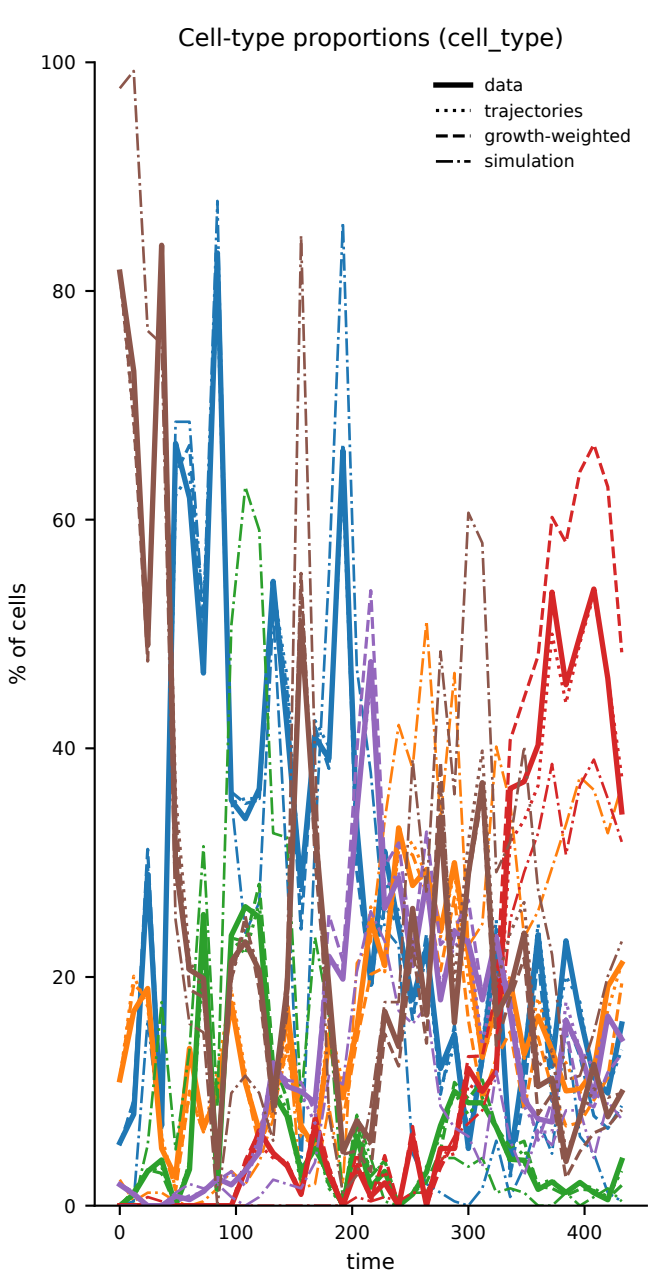


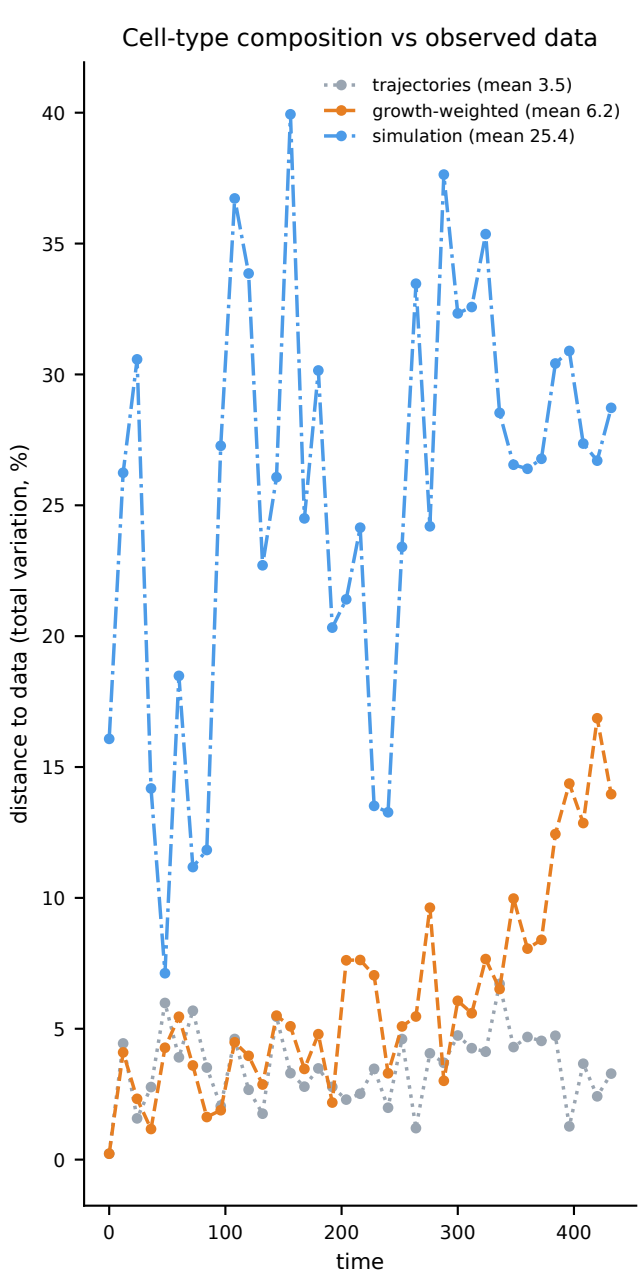


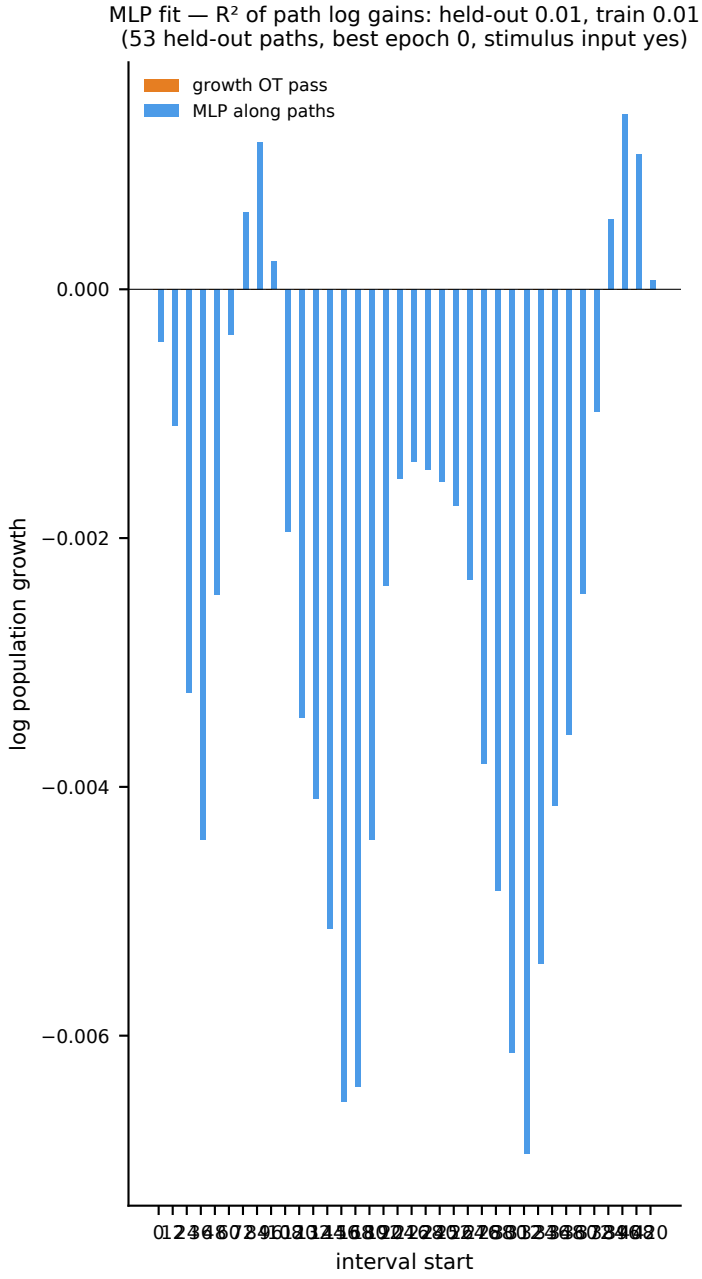
Neural Stromal Trophoblast iPSC Epithelial MEF

4. Proliferation — growth-weighted trajectories and MLP fit

The OT selection keeps one descendant per ancestor (trajectories without proliferation); weighting each path by its expansion exp(∫R_opt) should bring them back to the observed data. The simulation used the proliferation MLP.







Neural Stromal Trophoblast iPSC Epithelial MEF

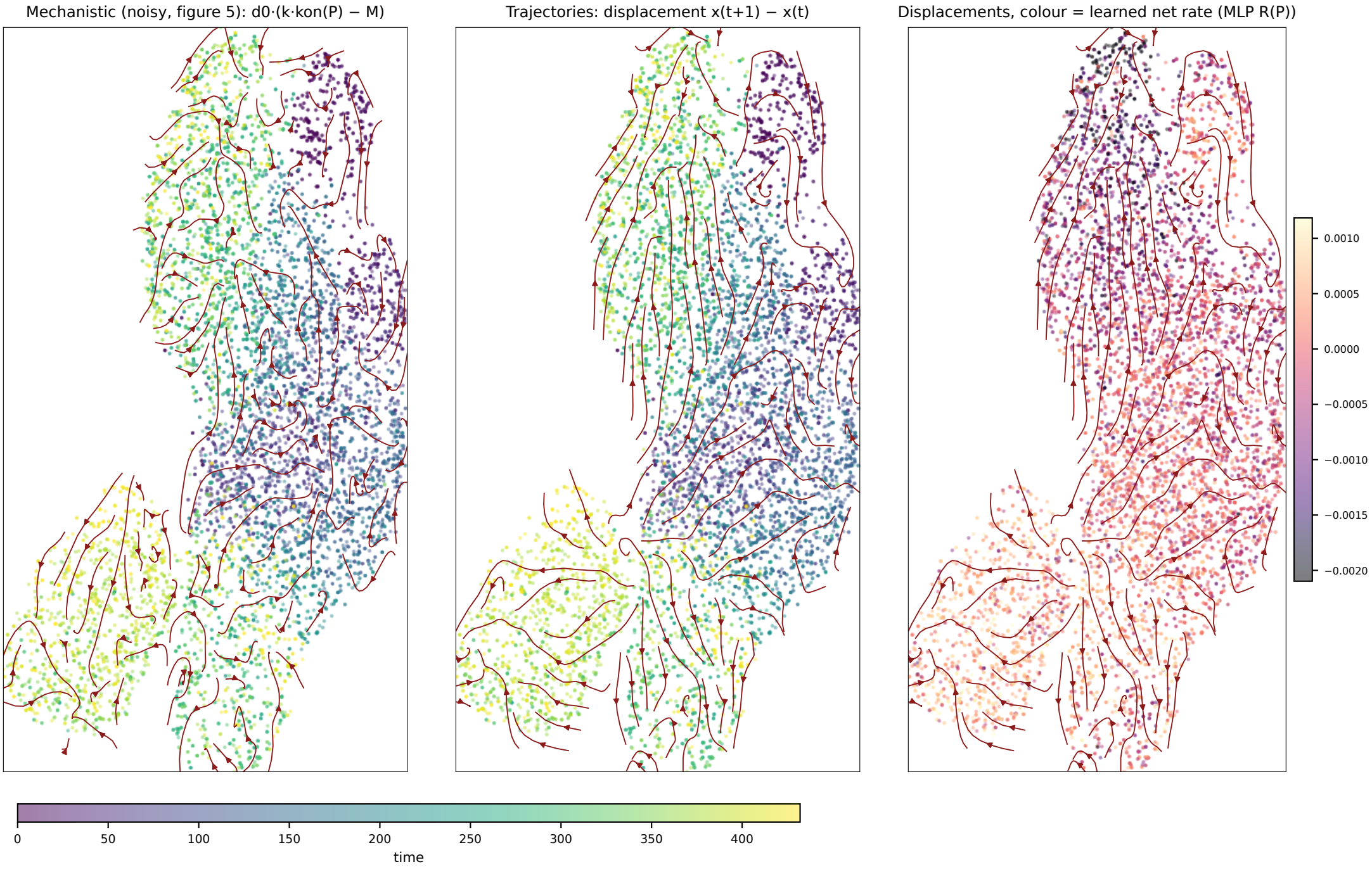
4. Protein dilution

Section unavailable:

```
ValueError('protein_dilution = False, or no birth rate (obs proliferation_birth_rate / net rate)')
```

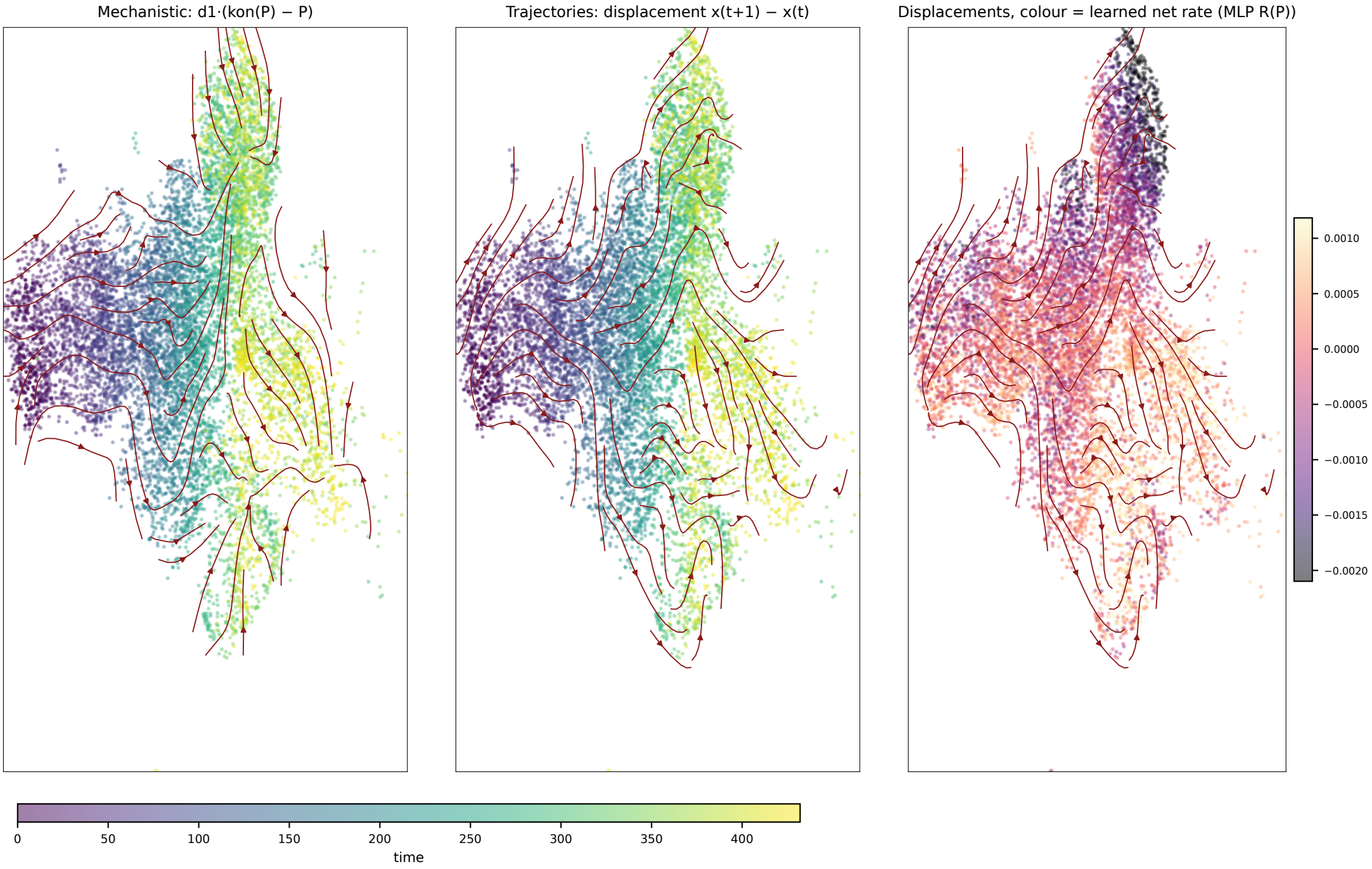
5. Learned dynamics — mRNA velocity and displacement fields

All 9768 trajectory states projected onto the UMAP learned on the observed mRNA (log1p, reference depth); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.10.



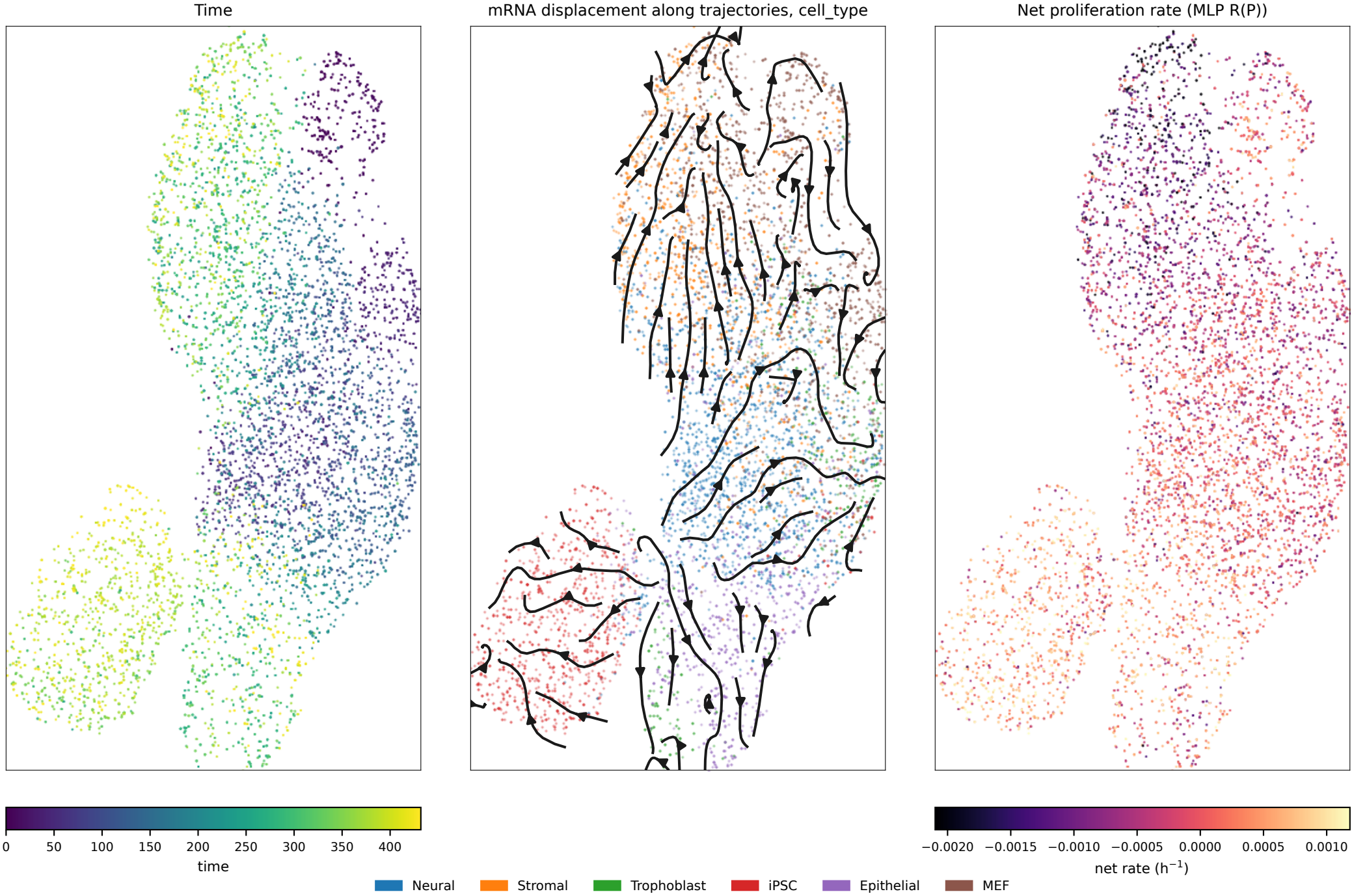
5. Learned dynamics — Proteins velocity and displacement fields

All 9768 trajectory states projected onto the UMAP learned on the protein trajectories (raw levels); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.48.



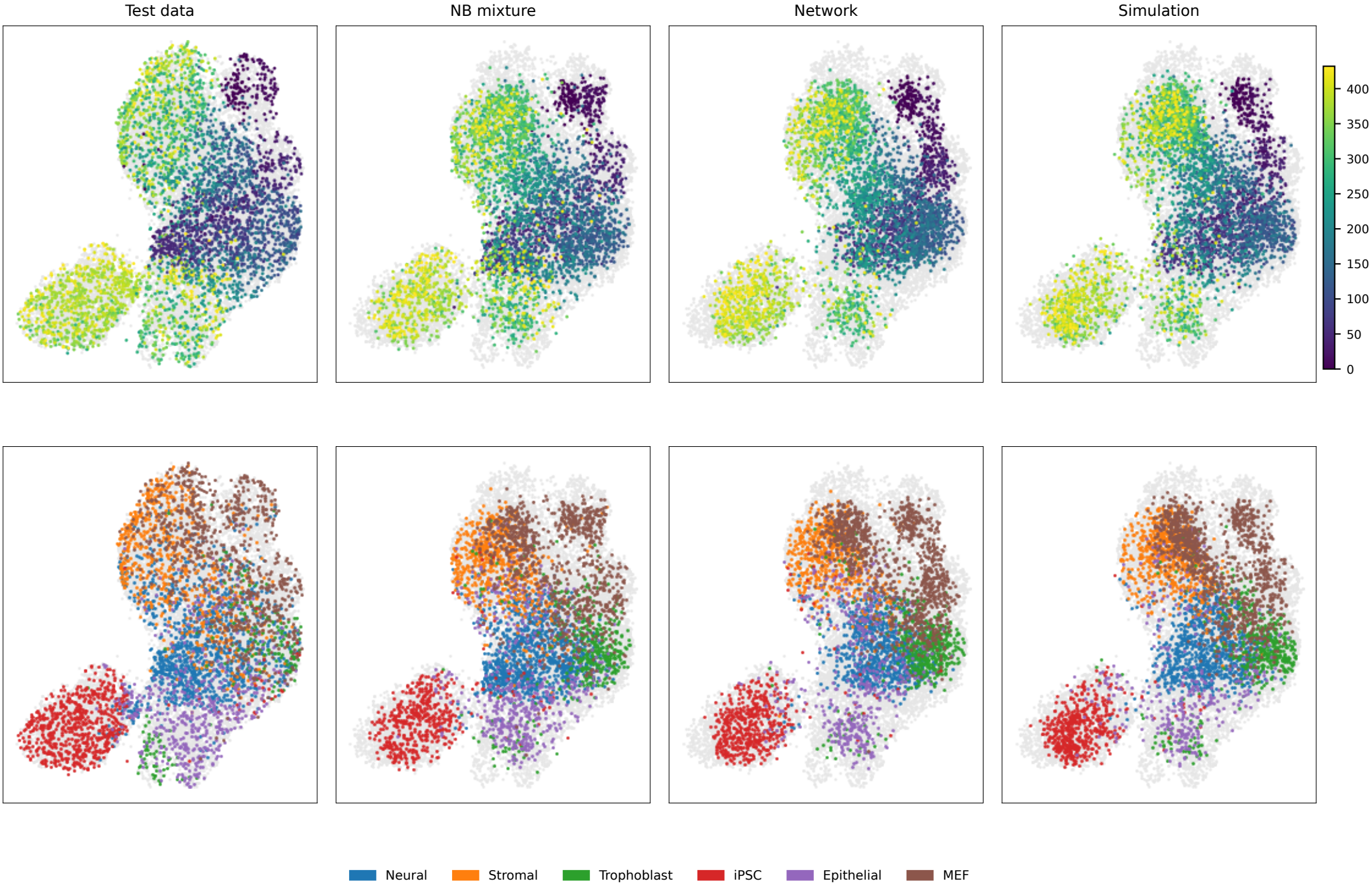
5. Learned dynamics — summary

All 9768 trajectory states on the UMAP learned on the observed mRNA (log1p, reference depth); displacement along the inferred trajectories x(t+1) − x(t); background of the field: cell_type.



6. Held-out test cells — predictions with the network fixed

Test cells classified into basins with the training mixtures, trajectories inferred with the training network fixed, then simulated; projected onto the UMAP learned on the observed data (grey: training).



6. Held-out test cells — quantitative checks

Per gene: 1-D Wasserstein distance between log1p counts, averaged over the common timepoints.

